

International Summer School on MS-based Metabolomics Data Processing

Tutorial session: 11:00 – 13:00

**Democratizing multi-omics data analysis
with BiomiX and Nextflow: from single-
omics to advanced integration**

Presenter:

Cristian IPERI, PhD

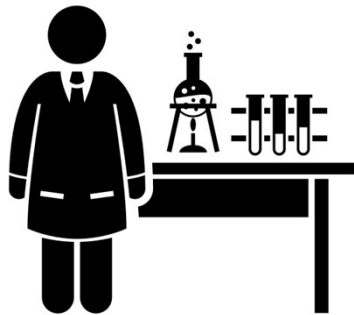
USZ: Center for Experimental
Rheumatology, Zurich Switzerland

cristian.iperi@uzh.ch

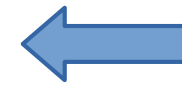


BIOMIX
MULTIOMICS DATA INTEGRATION

03-09-2026



**Access to data and
methods**



Clinicians, Specialists



Deep knowledge of the disease



Extract information from data

Bioinformaticians



Deep knowledge of the disease



Extract information from data



You have a biological question, but you are not a programmer



SEQUIN



**User-friendly tool for single omics
analysis existing but Fragmented**



**Multiomics integration tools with
accessible user interface**



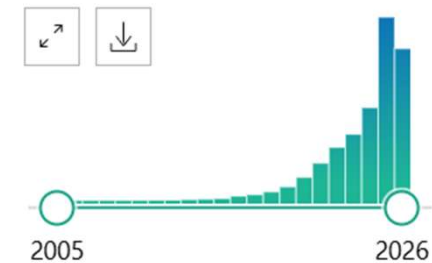
You have a biological question, but you are not a programmer



**User-friendly tool for single omics
analysis existing but Fragmented**

SEQUIN

RESULTS BY YEAR



Multiomics research on pubmed
12,031 articles on 24 June 2026.

**Multiomics integration tools with
accessible user interface**

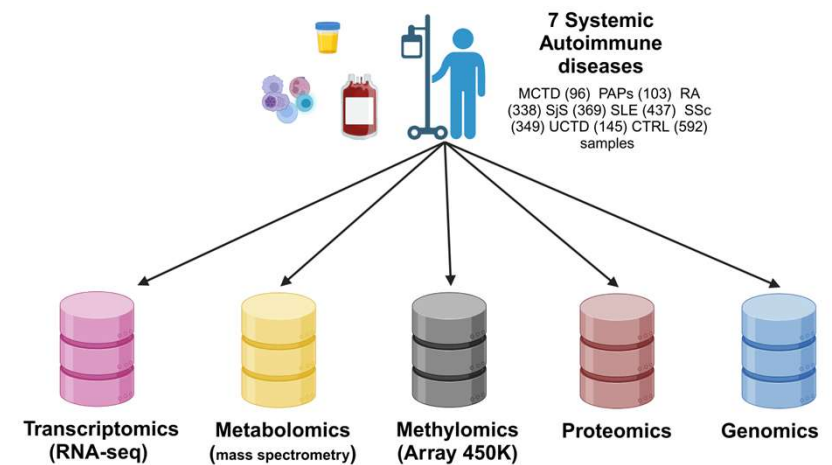


PRECISESADS

Molecular Reclassification to Find
Clinically Useful Biomarkers for
Systemic Autoimmune Diseases

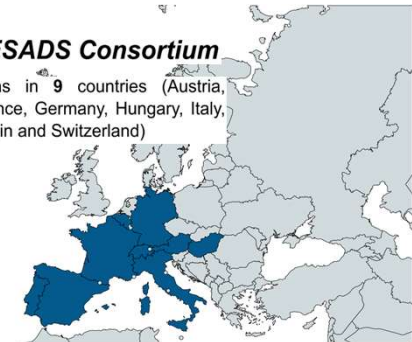
Rationale: Autoimmune disease clinical classifications show high heterogeneity within the disease.

Objective: Reclassification of the autoimmune diseases by biological and molecular criteria

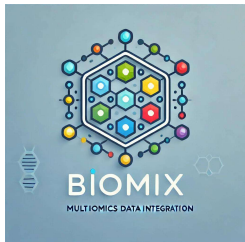
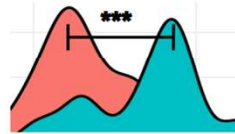
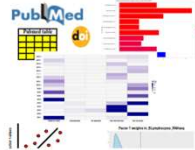
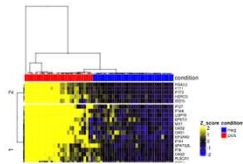
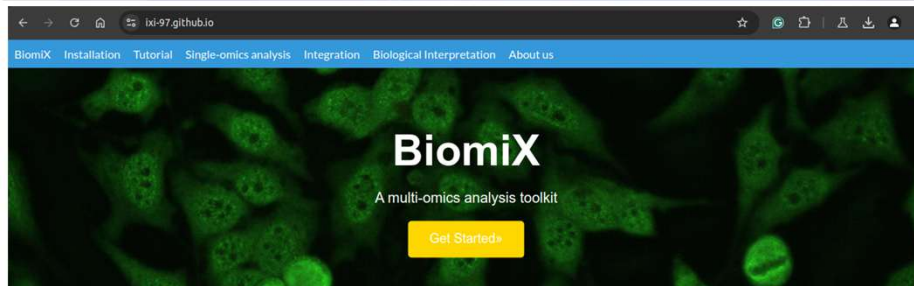


PRECISESADS Consortium

19 institutions in 9 countries (Austria, Belgium, France, Germany, Hungary, Italy, Portugal, Spain and Switzerland)



G. Barturen et al, Arthritis & Rheumatology (2021); L. Martin-Gutierrez, ACR (2021)



Software | [Open access](#) | Published: 10 January 2025

BiomiX, a user-friendly bioinformatic tool for democratized analysis and integration of multiomics data

[Cristian Iperi](#), [Álvaro Fernández-Ochoa](#), [Guillermo Barturen](#), [Jacques-Olivier Pers](#), [Nathan Foulquier](#), [Eleonore Bettacchioli](#), [Marta Alarcón-Riquelme](#), [PRECISESADS Flow Cytometry Study Group](#), [PRECISESADS Clinical Consortium](#), [Divi Cornec](#), [Anne Bordron](#) & [Christophe Jamin](#) 

BMC Bioinformatics 26, Article number: 8 (2025) | [Cite this article](#)

2834 Accesses | 1 Citations | 2 Altmetric | [Metrics](#)



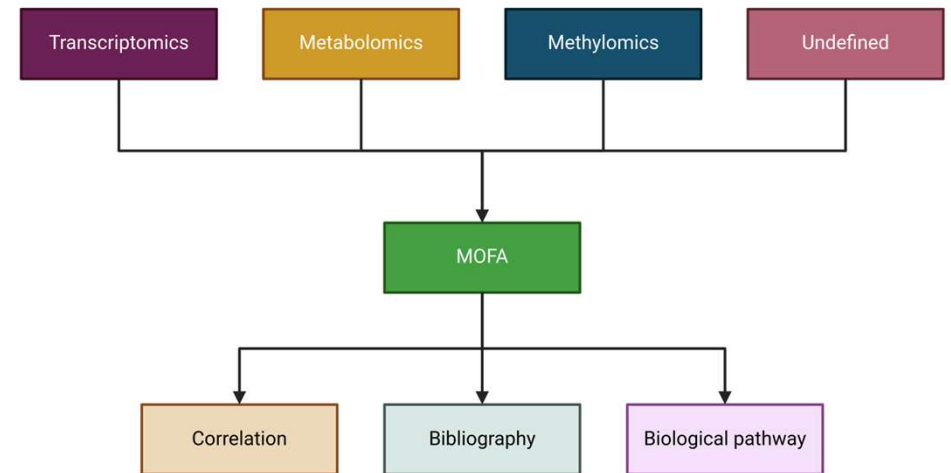
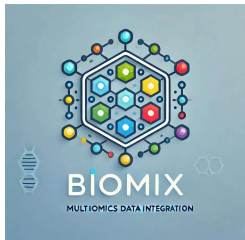
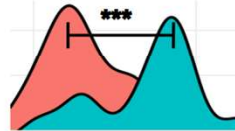
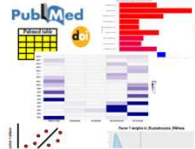
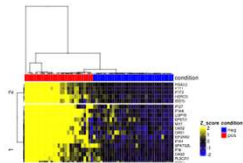
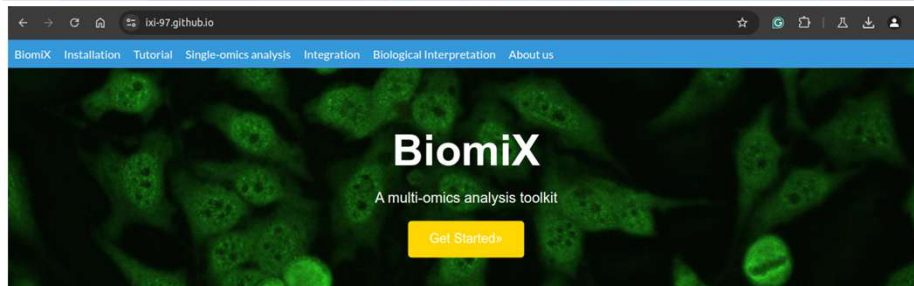
Website: <https://ixi-97.github.io/>



<https://github.com/ixi-97/BiomiX>



<https://ixi-97.github.io/Tutorial.html>



Website: <https://ixi-97.github.io/>



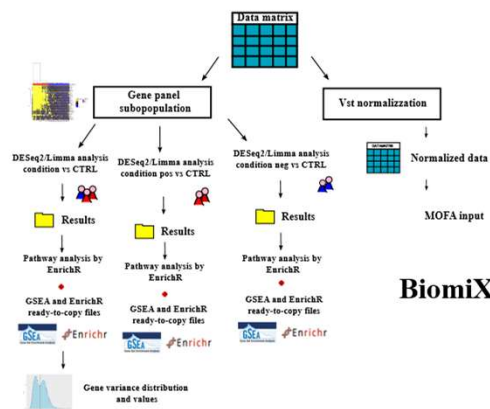
<https://github.com/ixi-97/Biomix>



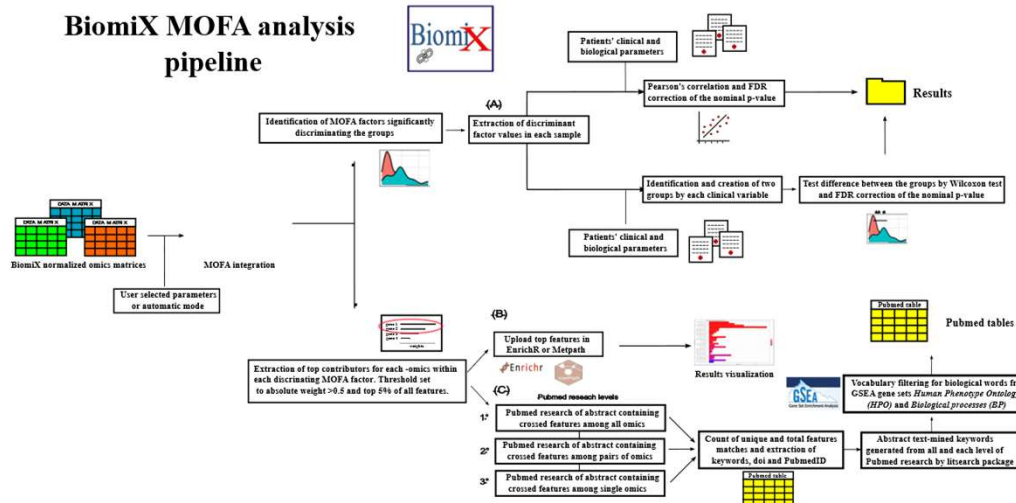
<https://ixi-97.github.io/Tutorial.html>



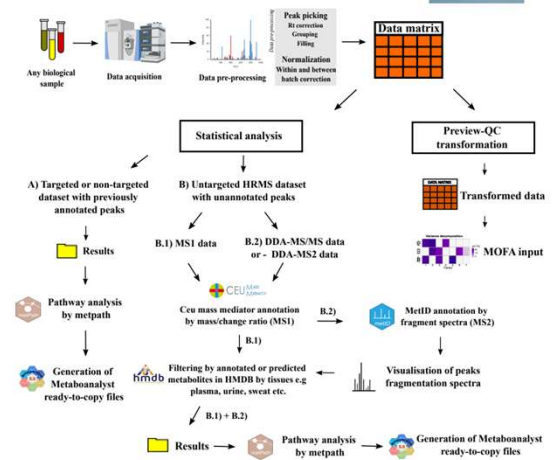
Biomix transcriptomics pipeline



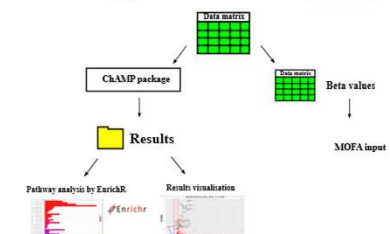
Biomix MOFA analysis pipeline



Biomix metabolomics pipeline



Biomix methylomics pipeline



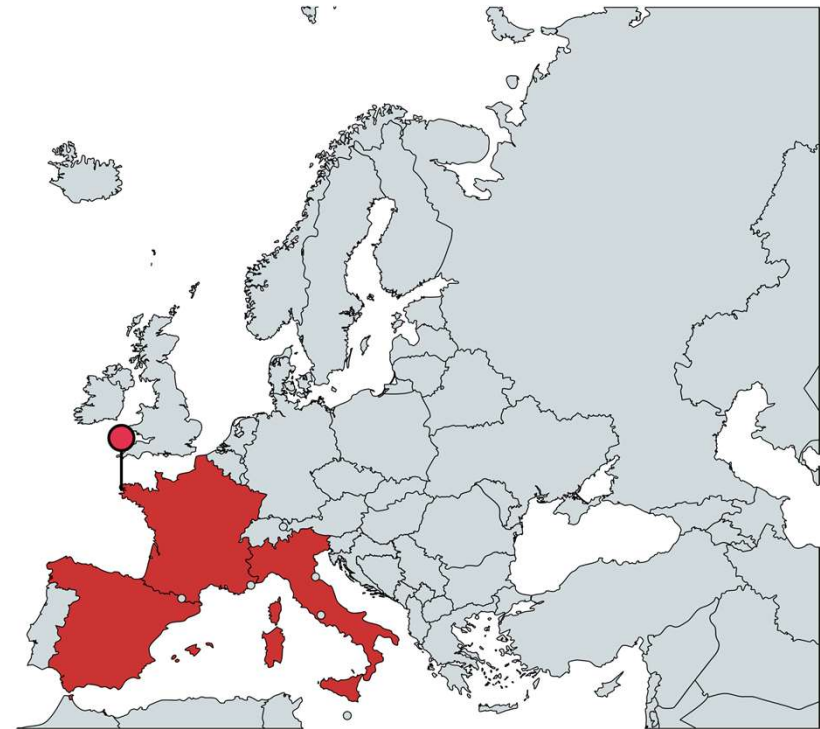
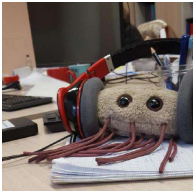


BiomiX, since then?

Hub of multiomics methods → **accessible to everyone**

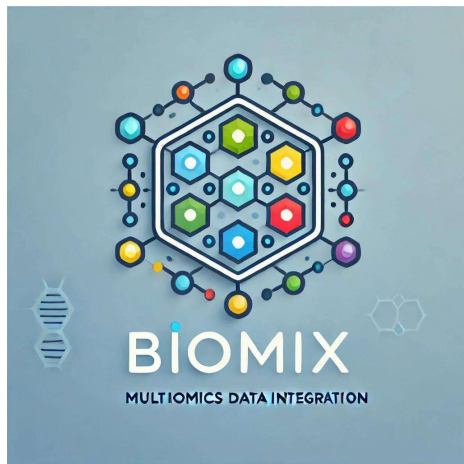
Increase the community of developers and users

Look at the **single and spatial omics** revolution

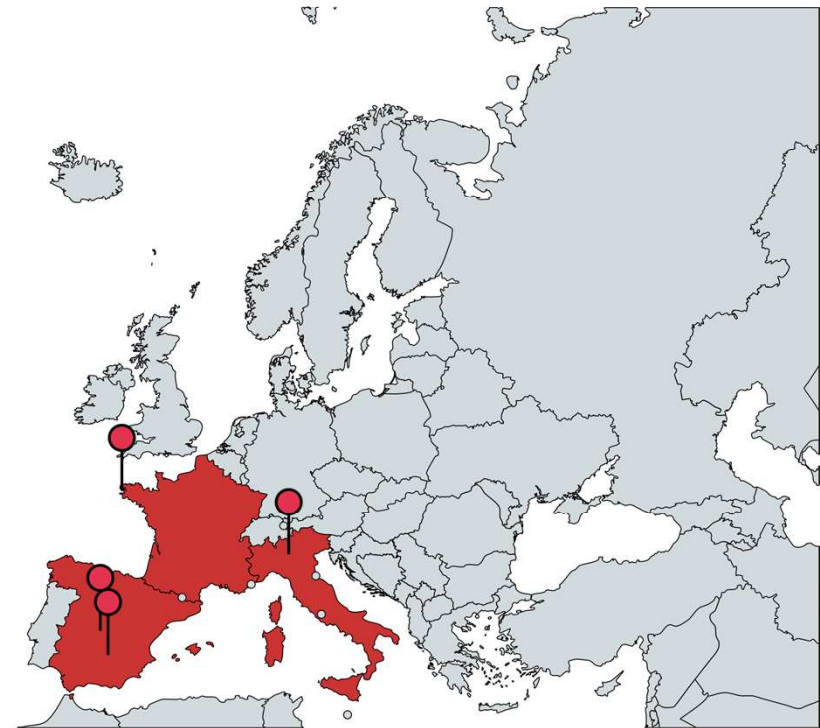


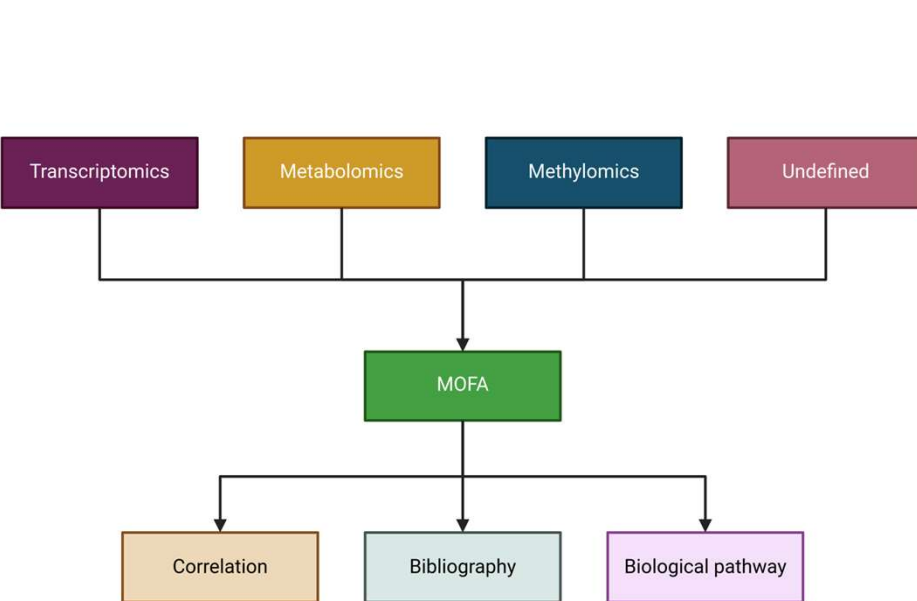


BiomiX, since then?

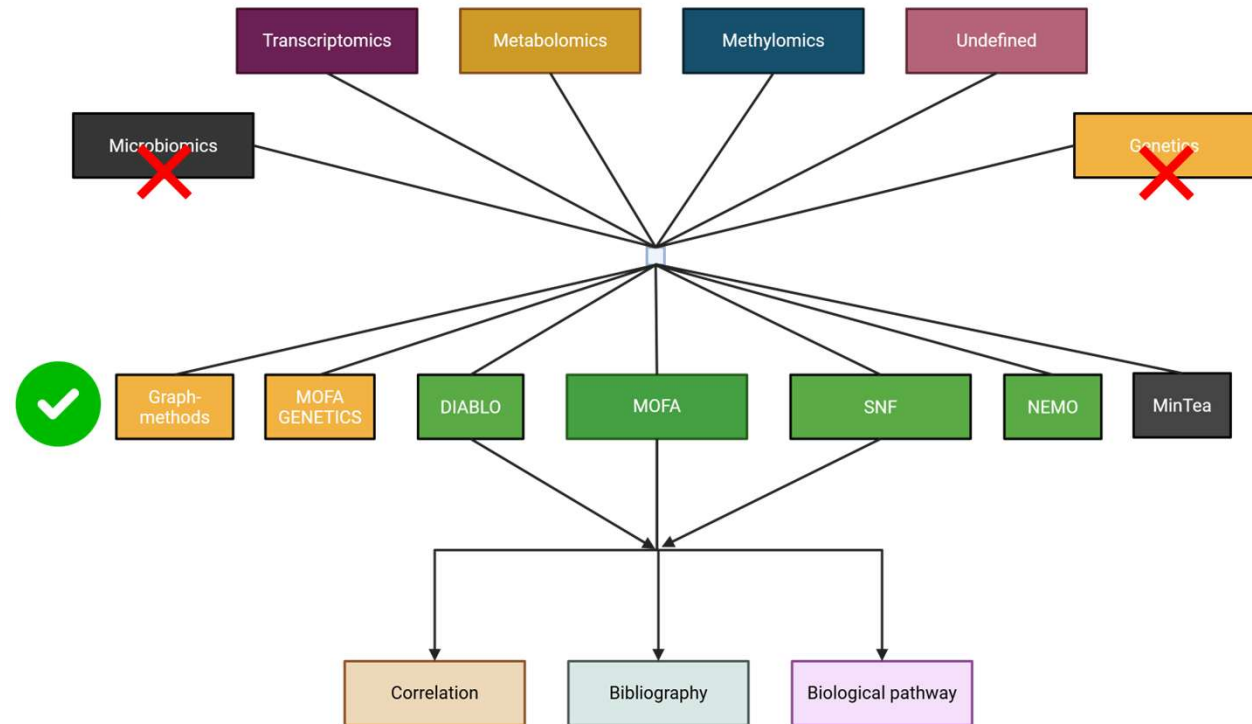


BiomiX consortium





BiomiX V2 2024



BiomiX 2026: Reality (V3.0)



Set up installation



BiomiX

3.0

[Home](#) [Installation](#) [Tutorial](#) [Single-omics](#) [Integration](#)

INSTALLATION

Get BiomiX 3.0 running

BiomiX 3.0 uses Docker to ensure a stable, reproducible environment across all platforms. The launcher scripts handle everything automatically, no command-line knowledge required.

0 Install Docker

BiomiX 3.0 requires Docker to be installed on your system. We recommend [Docker Engine](#) on Linux/Ubuntu, which is the environment BiomiX 3.0 has been built and tested for. New to Docker? Follow our [Docker installation guide](#).

A local installation without Docker is still possible using the legacy install scripts. See the [legacy installation section](#) at the bottom of this page for details.

Slide to be updated with the final version*

1) Open BiomiX website:
<https://biomix-consortium.github.io/>

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)



Set up installation



BiomiX 3.0

Home

Installation

Tutorial

Single-omics

Integration

Parameters

About

GitHub ↗

1 Download the BiomiX launcher

Download the launcher script for your operating system from the BiomiX GitHub repository. Place it in a folder of your choice, it will automatically create a `biomix_shared` folder nearby for your input data and results.



BiomiX_Windows.zip

Windows, contains .bat and .ps1



BiomiX_Start.sh

Linux, run from terminal



BiomiX-Launcher-1.0.0.dmg

macOS, open and install



All launchers are also available in the BiomiX GitHub repository under the `launchers/` folder.

Slide to be updated with the final version*

1) Open BiomiX website:
<https://biomix-consortium.github.io/>

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)

3) Download the BiomiX launcher



Set up installation

Slide to be updated with the final version*

2 Launch BiomiX

The launcher will automatically pull all required Docker modules on first run and open the BiomiX interface in your browser. Select your platform below.

Windows

Linux

macOS

1. Make sure **Docker Desktop is open and running** (look for the whale icon in the system tray).
2. **Double-click** `BiomiX_Start.bat`, a setup window will appear.
3. Choose the **folder** where your input data is located and where results will be saved.
4. Optionally, enter your **NCBI API key** to enhance PubMed search in the biological interpretation module. This is not required, BiomiX works fully without it.
5. Click **Start**. BiomiX will open automatically in your browser at `http://localhost:3838`.

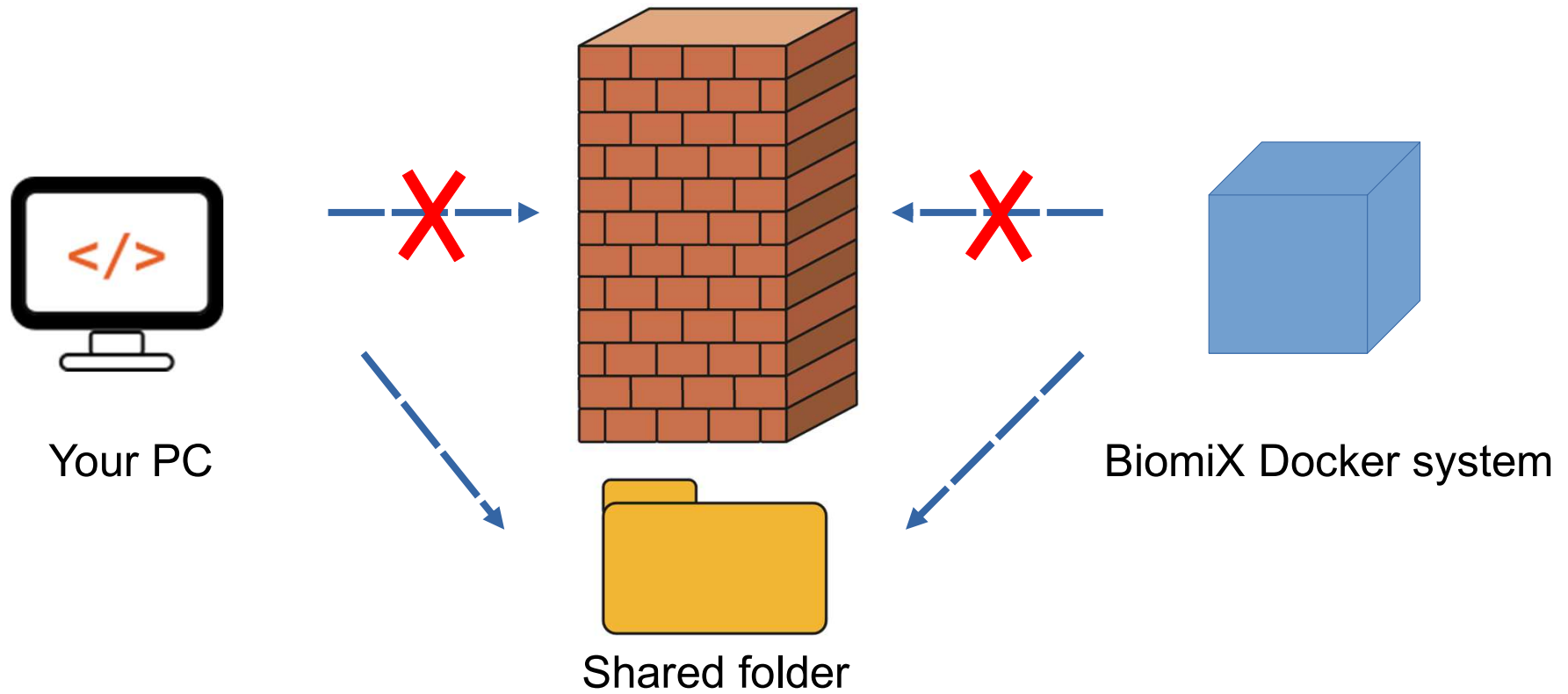


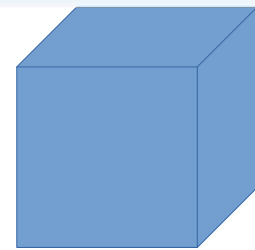
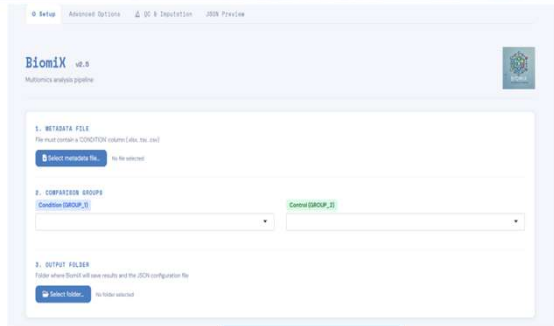
Keep the terminal window open while using BiomiX. Closing it will stop the application.

4) Launch BiomiX



How BiomiX works?

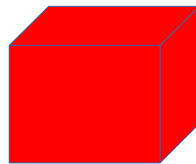




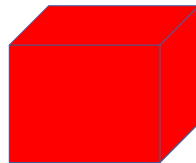
Biomix GUI
Docker



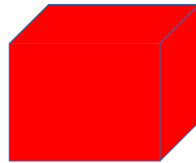
How Biomix works?



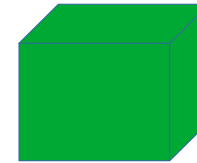
Docker transcriptomics



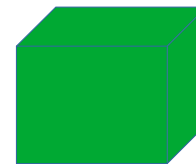
Docker metabolomics



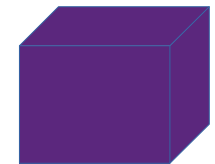
Docker methylomics



Docker MOFA/DIABLO



Docker SNF/NEMO



Docker Interpretation
• Pubmed/Clinical/Enrichment

Single omics dockers

Integration dockers

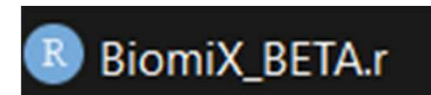
Interpretation dockers



How BiomiX works?

BiomiX interface

→ Json.file →



Master script coordinating
all the analysis.



Datasets of this tutorial

> *J Clin Invest*. 2018 Jan 2;128(1):427-445. doi: 10.1172/JCI93801. Epub 2017 Dec 11.

Drug-perturbation-based stratification of blood cancer

Sascha Dietrich^{1 2 3 4}, Małgorzata Oleś¹, Junyan Lu¹, Leopold Sellner^{2 3}, Simon Anders¹, Britta Velten¹, Bian Wu³, Jennifer Hüllelein^{1 3}, Michelle da Silva Liberio³, Tatjana Walther³, Lena Wagner³, Sophie Rabe^{1 2 3}, Sonja Ghidelli-Disse⁵, Marcus Bantscheff⁵, Andrzej K Oleś¹, Mikołaj Slabicki³, Andreas Mock¹, Christopher C Oakes^{6 7}, Shihui Wang³, Sina Oppermann³, Marina Lukas³, Vladislav Kim¹, Martin Sill⁸, Axel Benner⁸, Anna Jauch⁹, Lesley Ann Sutton¹⁰, Emma Young¹⁰, Richard Rosenquist¹⁰, Xiyang Liu³, Alexander Jethwa³, Kwang Seok Lee³, Joe Lewis¹¹, Kerstin Putzker¹¹, Christoph Lutz², Davide Rossi¹², Andriy Mokhir¹³, Thomas Oellerich^{14 15}, Katja Zirikli^{15 16}, Marco Herling¹⁷, Florence Nguyen-Khac¹⁸, Christoph Plass^{7 15}, Emma Andersson¹⁹, Satu Mustjoki¹⁹, Christof von Kalle^{3 15 20}, Anthony D Ho², Manfred Hensel²¹, Jan Dürig^{15 22}, Ingo Ringshausen²³, Marc Zapatka²⁴, Wolfgang Huber^{1 4}, Thorsten Zenz^{2 3 15 25}

Affiliations + expand

PMID: 29227286 PMCID: PMC5749541 DOI: 10.1172/JCI93801

> *Microbiol Spectr*. 2023 Aug 17;11(4):e0057723. doi: 10.1128/spectrum.00577-23. Epub 2023 Jul 31.

Integration of Metabolomics and Transcriptomics Reveals Major Metabolic Pathways and Potential Biomarkers Involved in Pulmonary Tuberculosis and Pulmonary Tuberculosis-Complicated Diabetes

Yunguang Wang¹, Xinxin He², Danna Zheng³, Qiang He⁴, Lifang Sun^{5 6}, Juan Jin^{2 4}

Affiliations + expand

PMID: 37522815 PMCID: PMC10434036 DOI: 10.1128/spectrum.00577-23

 98 CLL IGHV mutated  74 CLL IGHV unmutated  12 CLL IGHV unknown

 13 HC  13 PTB  13 PTB and DM

1) Chronic lymphocytic leukemia (CLL): EGAS00001001746

2) Pulmonary tuberculosis and diabetes complications: MTBLS7623



Datasets of this tutorial

> *J Clin Invest*. 2018 Jan 2;128(1):427-445. doi: 10.1172/JCI93801. Epub 2017 Dec 11.

Drug-perturbation-based stratification of blood cancer

Sascha Dietrich^{1 2 3 4}, Małgorzata Oleś¹, Junyan Lu¹, Leopold Sellner^{2 3}, Simon Anders¹, Britta Velten¹, Bian Wu³, Jennifer Hüllein^{1 3}, Michelle da Silva Liberio³, Tatjana Walther³, Lena Wagner³, Sophie Rabe^{1 2 3}, Sonja Ghidelli-Disse⁵, Marcus Bantscheff⁵, Andrzej K Oleś¹, Mikołaj Slabicki³, Andreas Mock¹, Christopher Marina Lukas³, Vladislav Kim¹, Martin Sill⁸, Axel Emma Young¹⁰, Richard Rosenquist¹⁰, Xiyang Joe Lewis¹¹, Kerstin Putzker¹¹, Christoph Lutz⁷, Thomas Oellerich^{14 15}, Katja Zirlik^{15 16}, Marco Christoph Plass^{7 15}, Emma Andersson¹⁹, Satu Anthony D Ho², Manfred Hensel²¹, Jan Dürig¹, Wolfgang Huber^{1 4}, Thorsten Zenz^{2 3 15 25}

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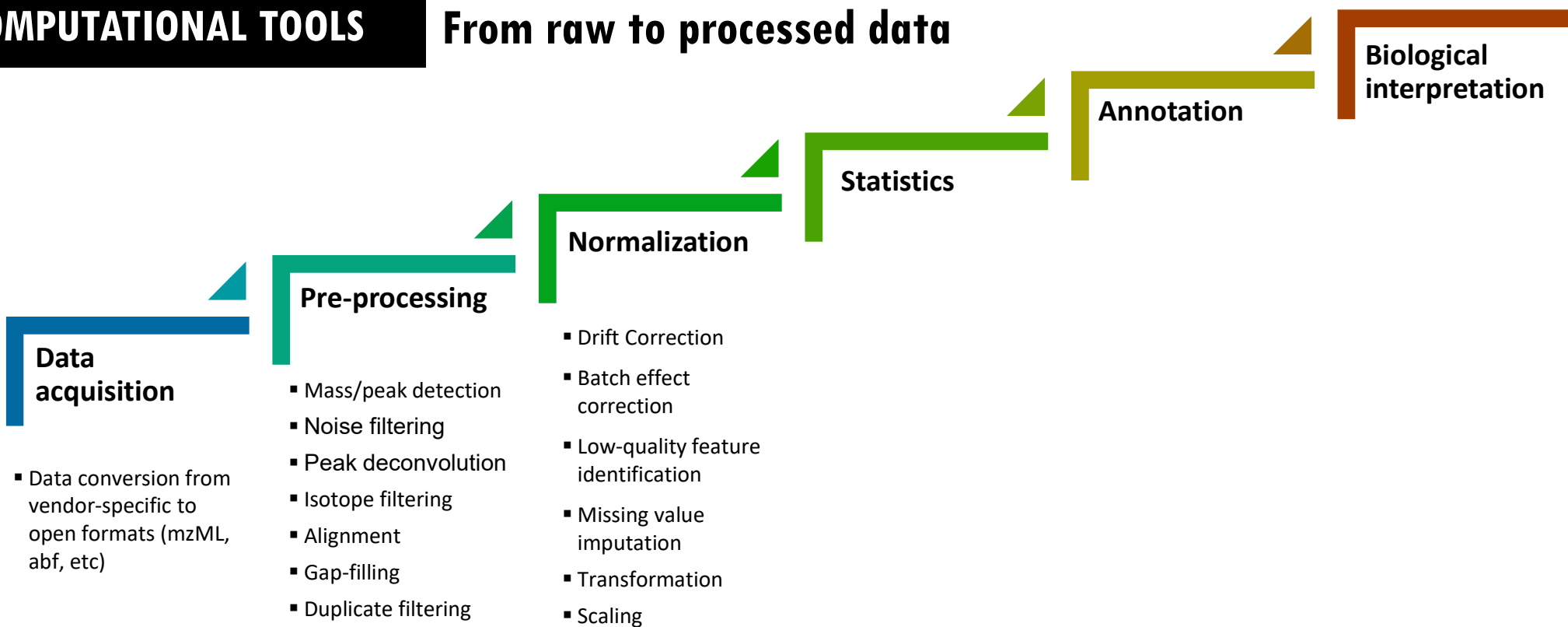
1) Chronic lymphocytic leukemia (CLL): EGAS00001001746

2) Pulmonary tuberculosis and diabetes complications: MTBLS7623



COMPUTATIONAL TOOLS

From raw to processed data





COMPUTATIONAL TOOLS

From raw to processed data

× Not covered by BiomiX

Data acquisition

- Data conversion from vendor-specific to open formats (mzML, abf, etc)

Pre-processing

- Mass/peak detection
- Noise filtering
- Peak deconvolution
- Isotope filtering
- Alignment
- Gap-filling
- Duplicate filtering

Normalization

- Drift Correction
- Batch effect correction
- Low-quality feature identification
- Missing value imputation
- Transformation
- Scaling

Statistics

Annotation

Biological interpretation

✓ Covered by BiomiX

Sophisticated QC-based methods (drift/batch-effect correction, feature filtering) are NOT covered by BiomiX in the current version

Transformation, scaling, NA imputation & several normalization methods: covered

BiomiX requires an already pre-processed data matrix as input.

Raw data pre-processing and QC-based drift/batch-effect correction must be performed beforehand, following proper QA/QC criteria (especially in untargeted metabolomics).



Data generation

Data
alignment/QC

Gene count

DGE analysis

Enrichment
analysis



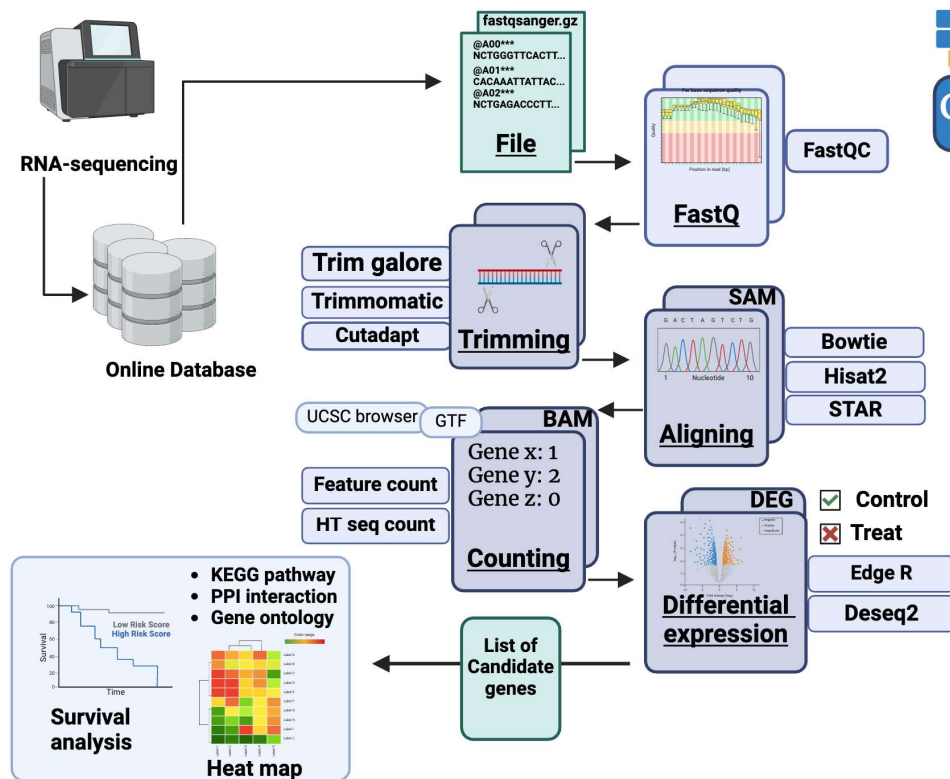
Not Supported by BiomiX
Cluster/High performance
computer running



BiomiX Supported!
Locally running



Transcriptomics analysis CLL

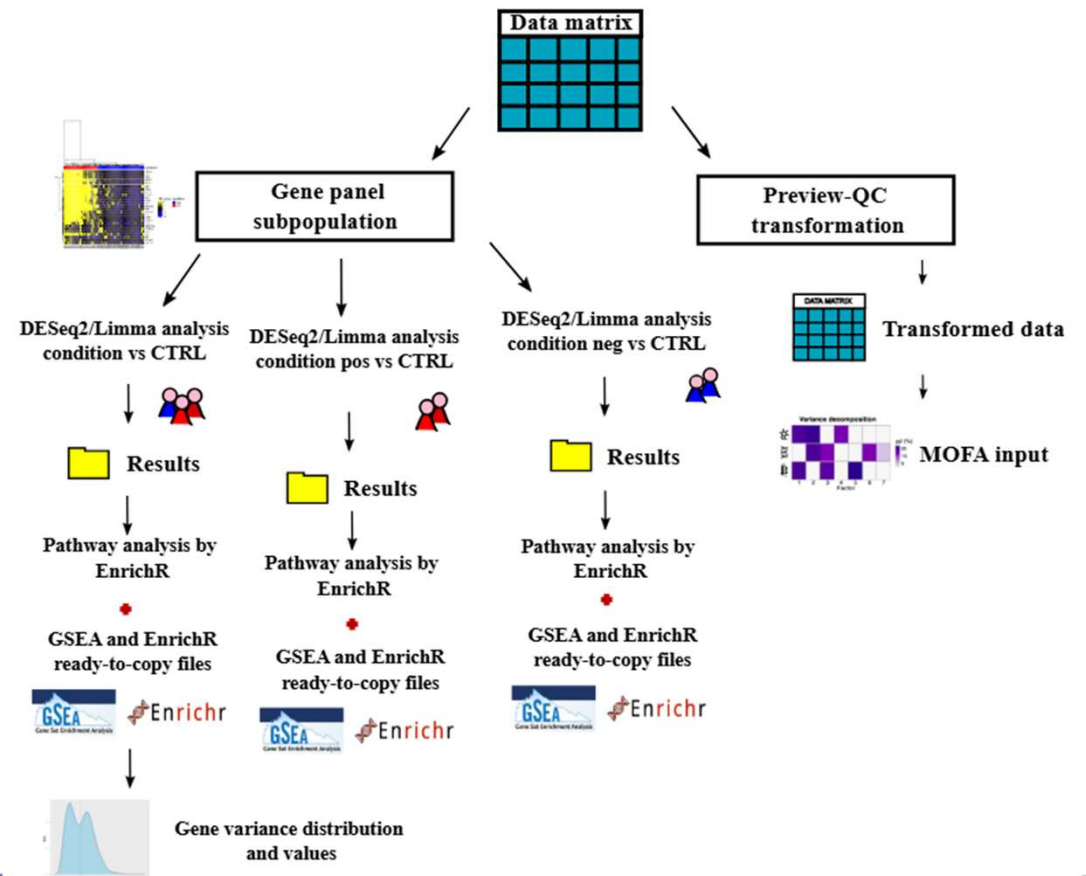
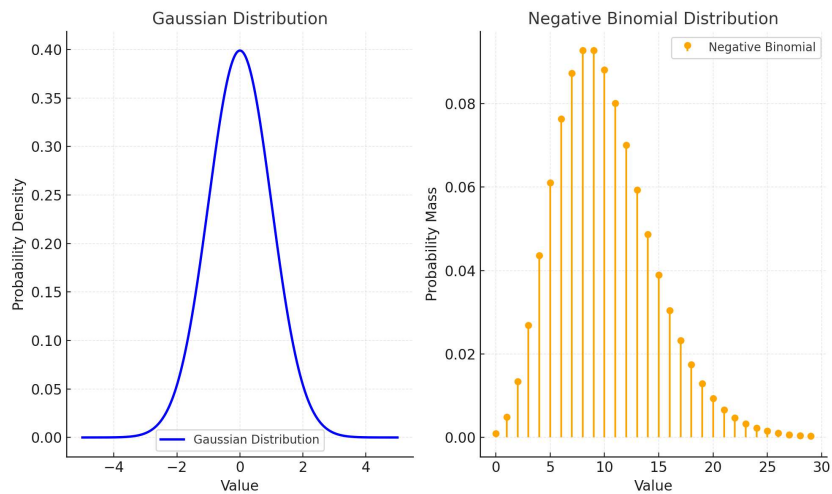
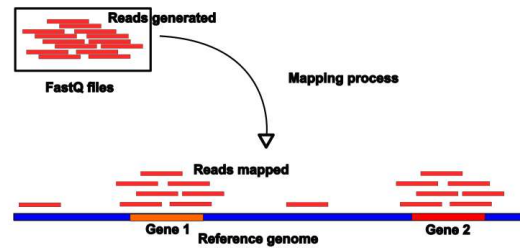


<https://usegalaxy.eu/>

Image adapted from: Sabbaghian A. (2024). User friendly procedure for conducting bulk RNA sequencing analysis using the GALAXY.eu platform. Biorender



BiomiX transcriptomics pipeline





Single-Omics tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



2026 Cristian Iperi, Jessica Gliozzo — CC BY 4.0



Transcriptomics analysis CLL

1) Open
BiomiX_Start.bat (Windows) **or**
BiomiX_Start.sh (Linux) **or**
BiomiX-Launcher-1.0.0.dmg (MacOS)

➔ Create a folder in Desktop called
“**shared**” and set it as BiomiX
reference directory.

BiomiX

Data folder (your input files and analysis results):

C:\Users\crist\Desktop\Shared Browse...

NCBI API key (optional - speeds up PubMed searches):

Get one for free at ncbi.nlm.nih.gov/account/settings

The first run will take longer while Docker downloads BiomiX components.

Cancel Start BiomiX



Transcriptomics analysis CLL

1) Load MTBLS7623_Metadata.tsv from Metadata folder.

Setup Advanced Options QC & Imputation JSON Preview

Biomix v2.5
Multiomics analysis pipeline



1. METADATA FILE
File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file... No file selected

2. COMPARISON GROUPS

Condition (GROUP_1)

Control (GROUP_2)

3. OUTPUT FOLDER
Folder where Biomix will save results and the JSON configuration file

Select folder... No folder selected



Transcriptomics analysis CLL

1) Load MTBLS7623_Metadata.tsv from Metadata folder.

Biomix v2.5
Multiomics analysis pipeline

1. METADATA FILE
File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file... No file selected

2. COMPARISON GROUPS
Condition (GROUP_1)

3. OUTPUT FOLDER
Folder where Biomix will save results and the JSON configuration file

Select folder... No folder selected

Select the metadata file

name	size	modified	created
Clinical_data		mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
COMMANDS.tsv	309 B	mercoledì 20 mag 2026, 08:37	mercoledì 4 giu 2025, 14:38
COMMANDS_ADVANCED.tsv	2.0 kB	mercoledì 15 apr 2026, 12:15	mercoledì 4 giu 2025, 14:38
COMMANDS_MOFA.tsv	134 B	mercoledì 20 mag 2026, 08:37	mercoledì 4 giu 2025, 14:38
directory.txt	74 B	martedì 9 giu 2026, 11:10	giovedì 5 giu 2025, 11:36
directory_out.txt	32 B	mercoledì 20 mag 2026, 08:36	giovedì 5 giu 2025, 17:40
Example_dataset		mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
Integration		lunedì 8 giu 2026, 09:01	mercoledì 4 giu 2025, 14:38
Metabolomics		martedì 11 nov 2025, 23:34	mercoledì 4 giu 2025, 14:38
Metadata		giovedì 7 mag 2026, 08:47	mercoledì 4 giu 2025, 14:38
Methylomics		martedì 11 nov 2025, 15:34	mercoledì 4 giu 2025, 14:38
modules		martedì 9 giu 2026, 17:06	venerdì 24 apr 2026, 13:26
Package_dependency_map.xlsx	18.7 kB	martedì 5 mag 2026, 08:34	martedì 5 mag 2026, 07:37
Report_parameters		martedì 9 giu 2026, 16:54	mercoledì 4 giu 2025, 14:38
Transcriptomics		martedì 11 nov 2025, 18:04	mercoledì 4 giu 2025, 14:38

Cancel Select



Transcriptomics analysis CLL

1) Load MTBLS7623_Metadata.tsv from Metadata folder.

Biomix v2.5
Multiomics analysis pipeline

1. METADATA FILE
File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)
[Select metadata file...](#) [MTBLS7623_Metadata.tsv](#)
39 samples · 3 conditions: HC, PTB, PTB_DM

2. COMPARISON GROUPS
Condition (GROUP_1) **Control (GROUP_2)**
PTB HC

3. OUTPUT FOLDER
Running in Docker: results will be saved to the folder you mounted on your machine via -v <your-folder>:/shared
/shared

Biomix need two metadata columns:
“ID”: Unique sample ID
“CONDITION”: Sample condition





OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined ▾	☐	Label input 1	☐	Browse file No file selected
Input 2	☐	☐	Undefined ▾	☐	Label input 2	☐	Browse file... No file selected
Input 3	☐	☐	Undefined ▾	☐	Label input 3	☐	Browse file... No file selected
Input 4	☐	☐	Undefined ▾	☐	Label input 4	☐	Browse file... No file selected
Input 5	☐	☐	Undefined ▾	☐	Label input 5	☐	Browse file... No file selected
Input 6	☐	☐	Undefined ▾	☐	Label input 6	☐	Browse file... No file selected

INTEGRATION OPTIONS

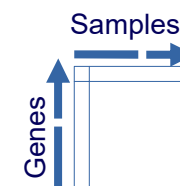
INTEGRATION
☐ Yes ☒ No

Browse file...

✓ EGAS00001001746_transcriptomics.xls

C:/Users/crist/Desktop/BioMiX2.5/Transcriptomics/INPUT/EG
AS00001001746_transcriptomics.xls

Upload: MTBLS7623_RNA_seq_OK.tsv



Slot system:
6 slot



Normalization and visualisation of the data before the analysis (Shiny app).

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined	☐	Label input 1	☐	Browse file... No file selected

Which omics? If not included select undefined (default)



Do you want to perform a single omics analysis or use the slot for the integration?

OMICS INPUTS							
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined	☐	Label input 1	☐	Browse file... No file selected



Personalized name for your dataset. If “filter” selected will only keep the sample sames containing this label.

	Lymphocytes_01	Monocytes_01
Gene 1		
Gene 2		
Gene 3		
Gene 4		
Gene 5		
Gene 6		
Gene 7		

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined ▾	☐	LYMPHOCYTES	☐	Browse file... No file selected



Run your first analysis

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Transcriptomics	☐	RNA	☐	Browse file... EGAS00001001746_transcriptomics.xls C:/Users/crist/Desktop/Biomix2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls
Input 2	☐	☐	Undefined	☐	Label input 2	☐	Browse file... No file selected
Input 3	☐	☐	Undefined	☐	Label input 3	☐	Browse file... No file selected

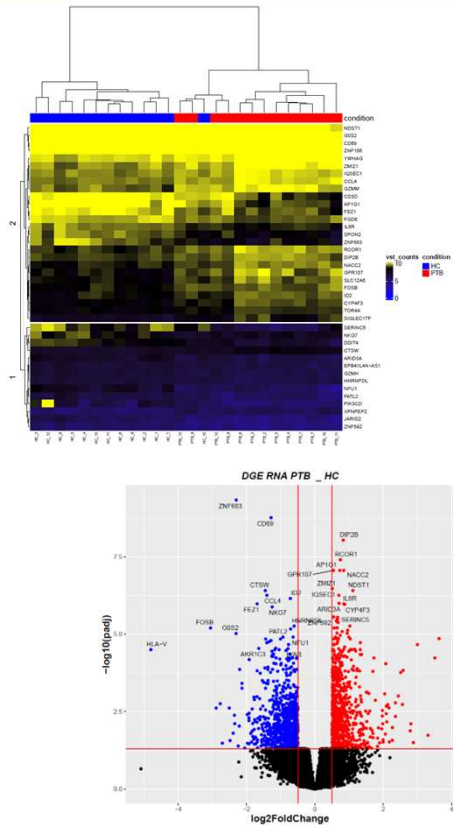
Advanced Options

Run Biomix

Download JSON

Ready to run

Check in Transcriptomics/Output folder → RNA_mutated_vs_unmutated created!



Input for GSEA program

Pathway_analysis	17/09/2026 16:10	Cartella di file	
Subpopulation_input	17/09/2026 16:10	Cartella di file	
Heatmap_top_genes_RNA_HC_vs_PT...	17/09/2026 18:21	Adobe Acrobat D...	16 KB
plot_DGE_PT...	17/09/2026 18:21	Adobe Acrobat D...	855 KB
PTB_HC_RNA.tsv	17/09/2026 18:21	File TSV	4.709 KB
PTB_HC_RNA_RAW.gct	17/09/2026 18:21	File GCT	11.652 KB
PTB_HC_RNA_RAW_meta.tsv	17/09/2026 18:21	File TSV	12 KB
PTB_HC_RNA_RAW_meta_groups.tsv	17/09/2026 18:21	File TSV	1 KB
RNA_PT...	17/09/2026 18:21	File TSV	7 KB
RNA_PT...	17/09/2026 18:21	File TSV	115 KB
RNA_PT...	17/09/2026 18:21	File TSV	93 KB
RNA_PT...	17/09/2026 18:21	File TSV	5 KB



DGE results all

	baseMean	log2FoldCI	lfcSE	stat	pvalue	padj
ENSG0000	1906.988	6.828504	0.533117	12.80864	1.4667914	4.56362816482032e-33
ENSG0000	6.345746	4.146017	0.349858	11.85056	2.1377006	3.32551409152136e-28
ENSG0000	111.8656	6.465554	0.559783	11.55012	7.3721385	7.64564487909312e-27
ENSG0000	386.3032	4.586128	0.413405	11.09356	1.3481534	1.04862742655686e-24

Only negative

Only Positive

Pathway_analysis	17/09/2026 16:10	Cartella di file	
Subpopulation_input	17/09/2026 16:10	Cartella di file	
Heatmap_top_genes_RNA_HC_vs_PTB_po...	17/09/2026 18:21	Adobe Acrobat D...	16 KB
plot_DGE_PTB_HC_RNA.pdf	17/09/2026 18:21	Adobe Acrobat D...	855 KB
PTB_HC_RNA.tsv	17/09/2026 18:21	File TSV	4.709 KB
PTB_HC_RNA_RAW.gct	17/09/2026 18:21	File GCT	11.652 KB
PTB_HC_RNA_RAW_meta.tsv	17/09/2026 18:21	File TSV	12 KB
PTB_HC_RNA_RAW_meta_groups.tsv	17/09/2026 18:21	File TSV	1 KB
RNA_PTB_HC_DOWN_ENRICH.R.tsv	17/09/2026 18:21	File TSV	7 KB
RNA_PTB_HC_GENESDOWN.tsv	17/09/2026 18:21	File TSV	115 KB
RNA_PTB_HC_GENESUP.tsv	17/09/2026 18:21	File TSV	93 KB
RNA_PTB_HC_UP_ENRICH.R.tsv	17/09/2026 18:21	File TSV	5 KB

Input for GSEA program



GSEA input



Enrichr

Input negative gene EnrichR

Input positive gene EnrichR

Pathway_analysis	17/09/2026 16:10	Cartella di file	
Subpopulation_input	17/09/2026 16:10	Cartella di file	
Heatmap_top_genes_RNA_HC_vs_PTB_po...	17/09/2026 18:21	Adobe Acrobat D...	16 KB
plot_DGE_PTB_HC_RNA.pdf	17/09/2026 18:21	Adobe Acrobat D...	855 KB
PTB_HC_RNA.tsv	17/09/2026 18:21	File TSV	4.709 KB
PTB_HC_RNA_RAW.gct	17/09/2026 18:21	File GCT	11.652 KB
PTB_HC_RNA_RAW_meta.tsv	17/09/2026 18:21	File TSV	12 KB
PTB_HC_RNA_RAW_meta_groups.tsv	17/09/2026 18:21	File TSV	1 KB
RNA_PTB_HC_DOWN_ENRICH.R.tsv	17/09/2026 18:21	File TSV	7 KB
RNA_PTB_HC_GENESDOWN.tsv	17/09/2026 18:21	File TSV	115 KB
RNA_PTB_HC_GENESUP.tsv	17/09/2026 18:21	File TSV	93 KB
RNA_PTB_HC_UP_ENRICH.R.tsv	17/09/2026 18:21	File TSV	5 KB



Enrichr

[Login](#) | [Register](#)

130,228,543 sets analyzed
473,654 terms
228 libraries

[Analyze](#)[What's new?](#)[Libraries](#)[Gene search](#)[Term search](#)[About](#)[Help](#)

Input data

Expand a gene, a term, or a variant into a gene set:



Try an example

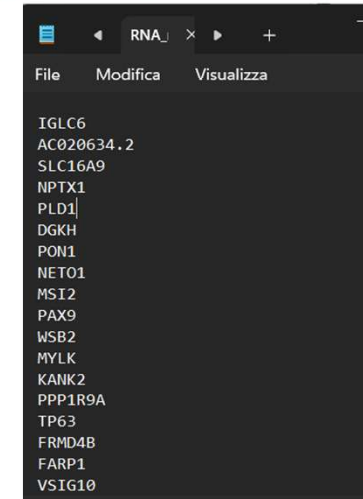
Include the top 100 most relevant genes



Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background**.

Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

0 gene(s) entered



Enrichr

[Login](#) | [Register](#)[Transcription](#) [Pathways](#) [Ontologies](#) [Diseases/Drugs](#) [Cell Types](#) [Misc](#) [Legacy](#) [Crowd](#)

Description No description available (100 genes)

Reactome Pathways 2024

CDG42 GTPase Cycle
RHOF GTPase Cycle
VEGFA-VEGFR2 Pathway
RHG GTPase Cycle
Signaling by GTPases

WikiPathways 2024

Human
Non Genomic Actions Of 1,25 Dihydroxyvita
Glycerolipids And Glycophospholipids WP
EGF EGFR Signaling WP437
Endothelin Pathways WP2197
Melanin Metabolism And Effects WP3298

BioPlanet 2019

GnRH signaling pathway
Delta Np63 pathway
PKC-catalyzed phosphorylation of inhibitory
EGF EGFR signaling pathway
Phosphatidylinositol signaling system



	A	B	C	D	E
1	#1.2				
2		52141	28		
3	NAME	Description	32140242	32140244	32141679
4	ENSG00000000003	NA	0	1.03138546799503	5.25875398402679
5	ENSG00000000005	NA	0	0	3.9440654880201
6	ENSG000000000419	NA	224.658295044468	207.308479067001	215.608913345099
7	ENSG000000000457	NA	411.72436542014	319.729495078459	337.874943473722
8	ENSG000000000460	NA	148.578792738572	86.6363793115826	101.231014192516
9	ENSG000000000938	NA	2134.7013294066	1614.11825741222	6782.47795089856
10	ENSG000000000971	NA	6.26537077813257	2.06277093599006	2.6293769920134
11	ENSG000000001036	NA	107.406356196558	111.389630543463	157.762610520904
12	ENSG000000001084	NA	247.929672220389	390.895092370117	210.350159361072
13	ENSG000000001167	NA	351.755816543728	257.846366998758	269.511141681373
14	ENSG000000001460	NA	48.3328602884512	45.3809605917813	74.9372442723818
15	ENSG000000001461	NA	508.390085997042	450.715449513828	590.295134707008
16	ENSG000000001497	NA	412.619418388445	414.616958134002	395.721237298016
17	ENSG000000001561	NA	239.874195505647	369.235997542221	437.791269170231
18	ENSG000000001617	NA		0.309415640398509	3.9440654880201
19	ENSG000000001626	NA		0.103138546799503	9.20281947204689
20	ENSG000000001629	NA	848.51021395281	877.709033263771	1020.1982729012
21	ENSG000000001630	NA	332.959704209331	354.796600990291	499.581628482546

Expression matrix in .gct format



Metadata_gsea

GSEA 4.1.0 (Gene set enrichment analysis)

File Downloads Help

Steps in GSEA analysis

- Load data
- Run GSEA
- Leading edge analysis
- Enrichment Map Visualization
- Tools
 - Run GSEAPreranked
 - Collapse Dataset
 - Chip2Chip mapping
- Analysis history

Home Run Gsea x Load data x

Gsea: Set parameters and run enrichment tests

Required fields

Expression dataset 

Gene sets database [//pub/gsea/gene_sets/c2.cp.kegg.v2023.1.Hs.symbols.gmt](http://pub/gsea/gene_sets/c2.cp.kegg.v2023.1.Hs.symbols.gmt)

Number of permutations 1000

Phenotype labels 

Collapse/Remap to gene symbols Collapse

Permutation type phenotype

Chip platform rsioned/Human_Ensembl_Gene_ID_MSigDB.v2023.1.Hs.chip

Basic fields Show

Advanced fields Show

16:31:45 61M of 124M

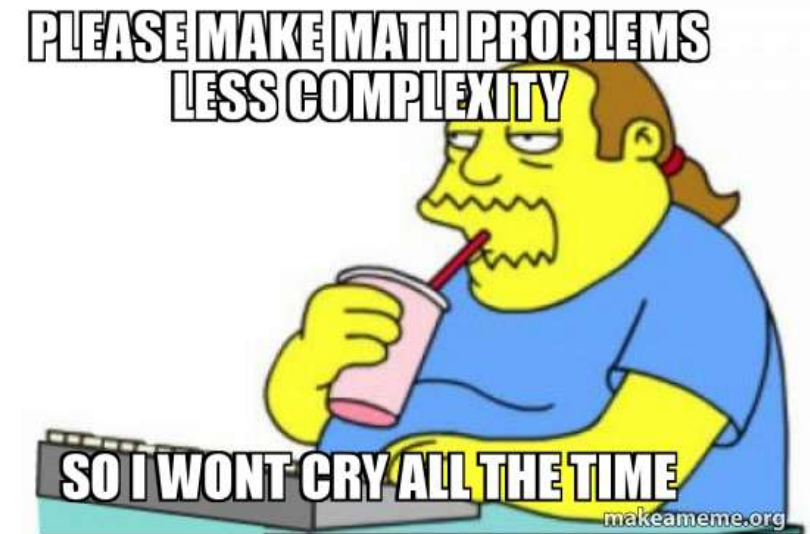


Analysis customization

Just running default parameters can not be sufficient,

What if I want

- Increase the log₂FC threshold
- Select only Male patients
- Stratify my patients by IFN signature





Analysis customization

Advanced Options

Run BiomiX

Download JSON

- Log2 fold change
- Pvalue
- N* gene in the heatmap
- Sample selection by gene panel

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

TRANSCRIPTOMICS OPTIONS

Log2FC threshold 0,5 Padj threshold 0,05 Gene panel file (optional) Browse... No file (optional)

METABOLOMICS OPTIONS

Log2FC threshold 0,5 Padj threshold 0,05

METHYLOMICS OPTIONS

Padj threshold 0,05 Log2FC threshold 0,15 Array type 450K

GENE PANEL AND POSITIVITY CRITERIA

N* genes Z-score > 2 10 N* genes Z-score > 1 10 N* top DE genes in heatmap 20 Subgrouping ☐ Enable subgrouping

CLUSTERING OPTIONS

Clustering distance euclidean Clustering method ward.D2

COMPUTATIONAL RESOURCES / GENERAL INTEGRATION OPTIONS

CPU threads 3 N* input features for integration 5000



Analysis customization

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo SNF / NEMO MintTea **Metadata Filtering**

❗ Column choices are populated automatically from the metadata file loaded in the Setup screen. Load the metadata file first if the dropdowns are empty.

SAMPLE FILTERING RULES

	COLUMN	DATA TYPE	THRESHOLD / VALUE	
F1	GENDER	Categorical	== male	☒
F2	ID	ID	e.g. 50 or GroupA	☐
F3	ID	ID	e.g. 50 or GroupA	☐
F4	ID	ID	e.g. 50 or GroupA	☐

▼ 1 active filter(s):

- Filter 1: GENDER [Categorical] == male

Categorical
== equal to
!= different to

Numerical
> (>=) major of
< (<=) minor of
== equal to

Always space*



Other functions?

OMICS INPUTS							
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☒	☒	Undefined	☐	Label input 1	☐	Browse file... No file selected



Normalization and interactive visualisation of the data before the analysis (Shiny app).

⚙️ Advanced Options

 **▶ Run Biomix**

 Download JSON

 Ready to run



Same structure, different omics

Advanced Data Normalization and Visualization

Plot Data Table Outliers Download

Select X Variable:
ENSG00000211899

Select Y Variable:
ENSG00000198804

Update Scatter Plot

Transformation methods

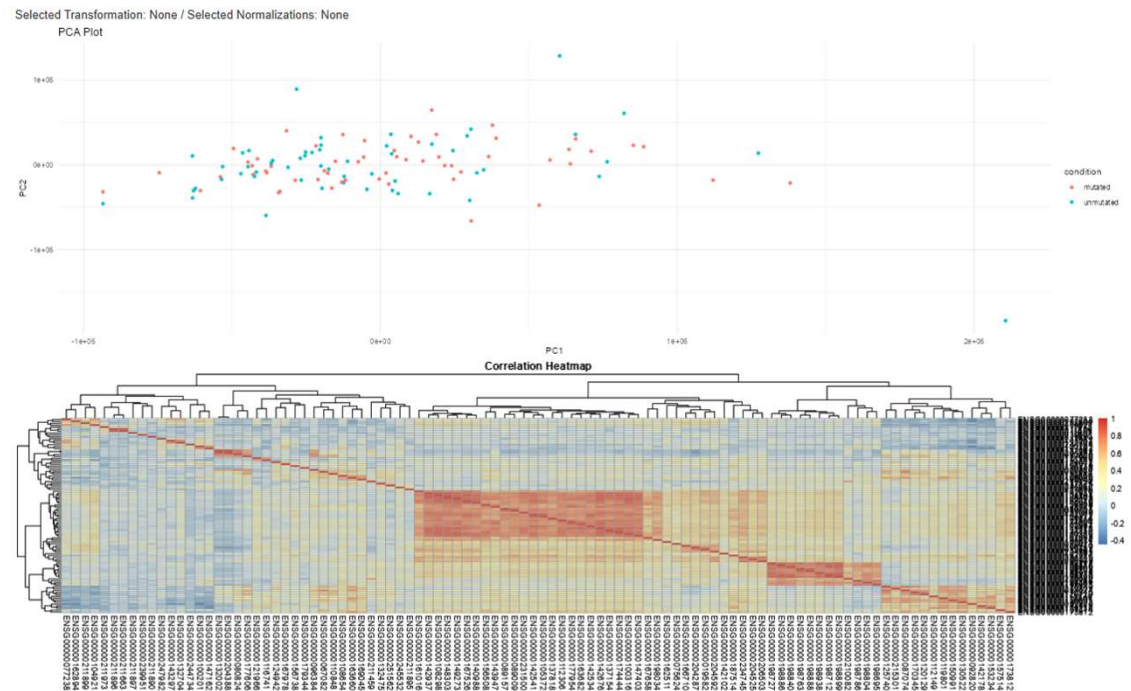
☒ Logarithmic
☐ Median Centralization
☐ Mean Centralization

Normalization methods

☐ Z-Score
☐ MAD
☐ Quantile Normalization
☐ Loess Normalization
☐ VST (Variance Stabilizing Transformation)

Choose Column for Coloring:
condition

Apply Normalization Reset to Original Data Download Current Plot Close App





Same structure, different omics

Advanced Data Normalization and Visualization

Plot Data Table Outliers Download

Select X Variable:
ENSG00000211899

Select Y Variable:
ENSG00000198804

Update Scatter Plot

Transformation methods

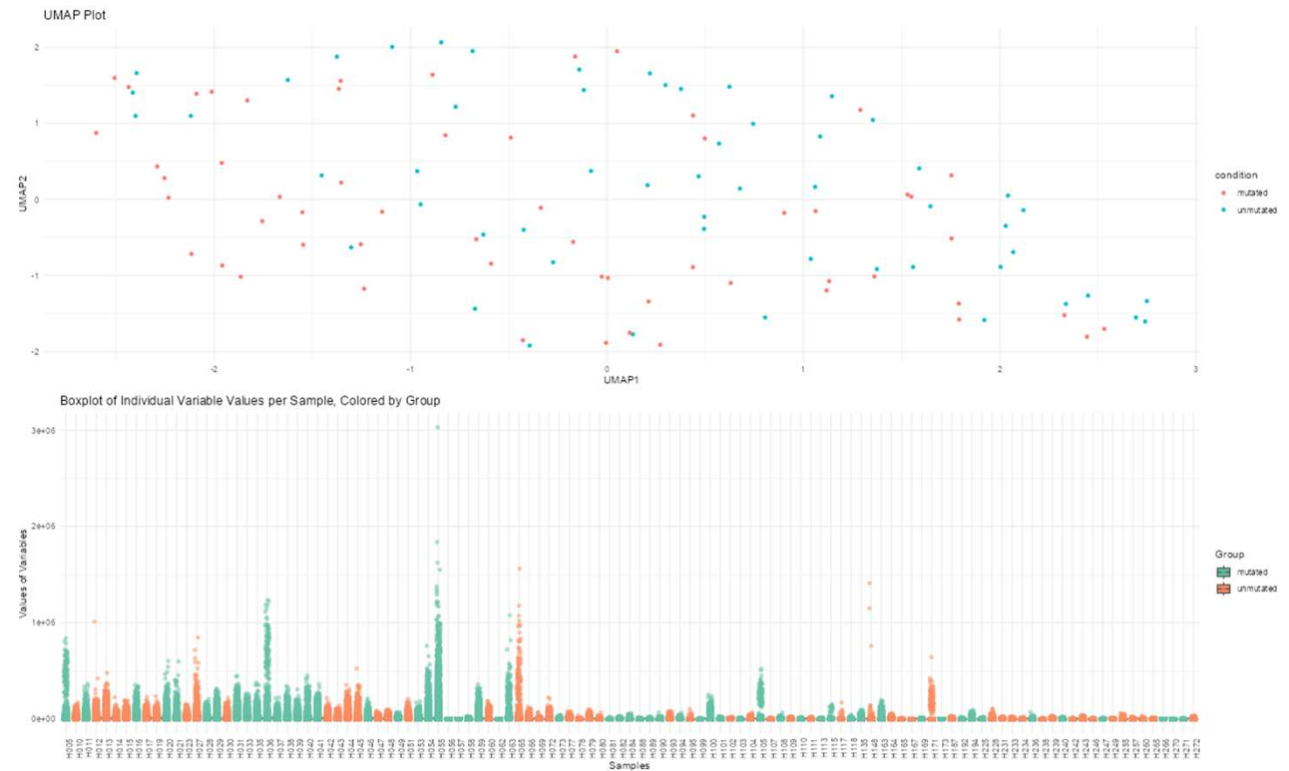
☒ Logarithmic
☐ Median Centralization
☐ Mean Centralization

Normalization methods

☐ Z-Score
☐ MAD
☐ Quantile Normalization
☐ Loess Normalization
☐ VST (Variance Stabilizing Transformation)

Choose Column for Coloring:
condition

Apply Normalization Reset to Original Data Download Current Plot Close App





Other functions?

Setup Advanced Options **QC & Imputation** JSON Preview

QC & IMPUTATION

Choose Imputation Method:

Replace with 0

Variable Missingness Threshold (%)

50

Sample Missingness Threshold (%)

50

Impute the matrix

Filter Variables and Samples

Reset to Original Data

Upload Data

Boxplot by Variables

Barplot by Samples

Summary Table

Data Table

Download

Upload a File

Browse... No file selected

☐ Transpose Data

☐ Remove First Row

☐ Remove First Column

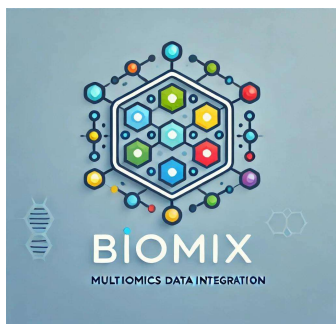
Load Data

QC & Imputation

- Imputation of missing data
- Removal of sample and variables with NaN data
- Save of the new table

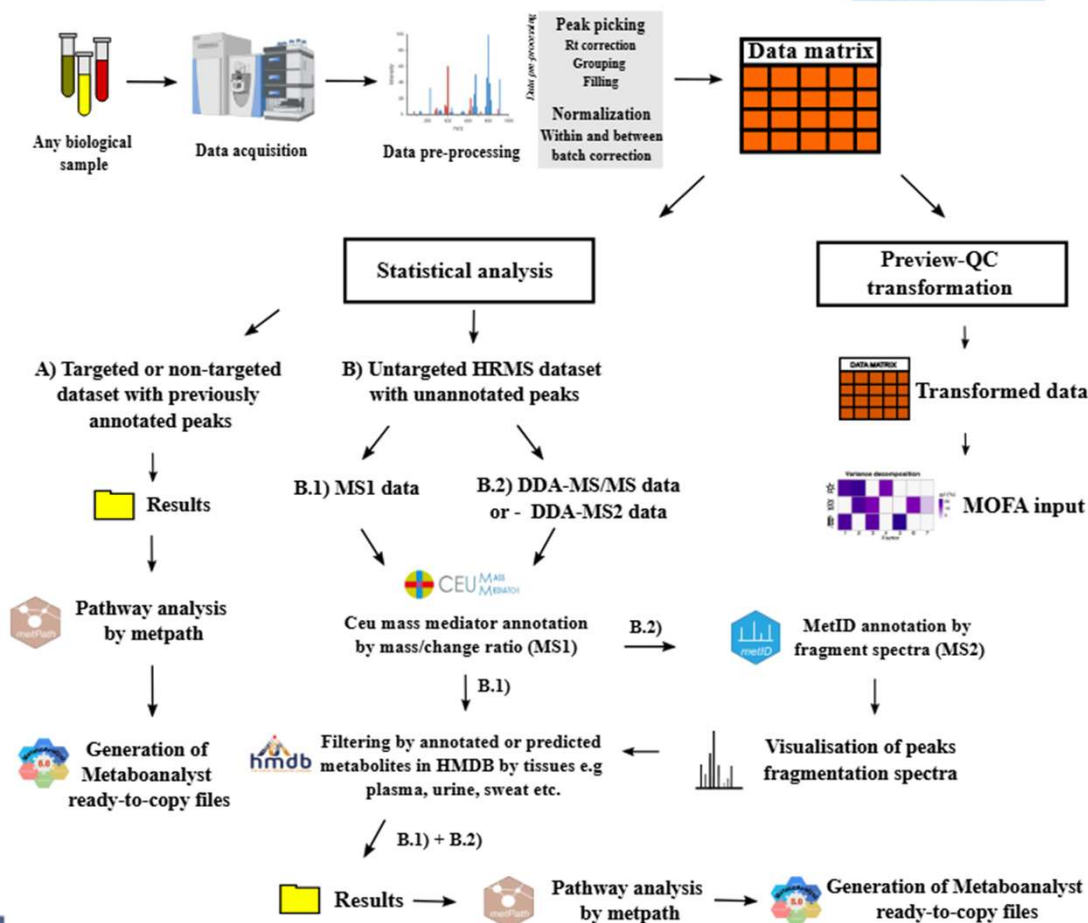


Metabolomics analysis PTB



UNIVERSIDAD
DE GRANADA

BiomiX metabolomics pipeline





1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

MTBLS7623_Metadata.xls

39 samples · 3 conditions: HC, PTB, PTB_DM

2. COMPARISON GROUPS

Condition (GROUP_1)

PTB

Control (GROUP_2)

HC

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

Select folder...

C:/Users/crist/Desktop/Biomix2.5

1) Load MTBLS7623_Metadata.xlsx from Metadata folder. (Metadata slot)

2) Load MTBLS7623_positive_mode_annotated_HMDB.xls from Metabolomics folder. (Input 1)

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION (•RUN SINGLE OMICS BEFORE)	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Transcriptomics	☐	RNA	☐	Browse file... MTBLS7623_RNA_seq_OK.tsv /shared/MTBLS7623_RNA_seq_OK.tsv
Input 2	☐	☒	Metabolomics	☐	METAB	☐	Browse file... MTBLS7623_positive_mode_annotated_HMDB.tsv /shared/MTBLS7623_positive_mode_annotated_HMDB.tsv

	A	B	C	D
1	ID	HC_1	HC_10	HC_11
2	HMDB0012497	8958855	11731.87022	11096.62856
3	HMDB0000925	3062.01725	2260.162667	3635.334667
4	HMDB0034301	174354.5458	136631.2239	116859.7333
5	HMDB0034301	50937.20491	44507.57589	50626.45636
6	HMDB0031666	9584.321591	10557.13338	9649.629818
7	HMDB0000097	81410.99886	82769.48167	68992.30848



Metabolomics analysis PTB

⚙ Setup

Advanced Options

🔬 QC & Imputation

JSON Preview

General

Metabolomics

MOFA & Diablo

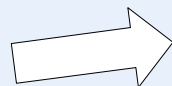
SNF / NEMO

MintTea

Metadata Filtering

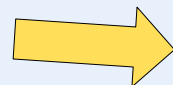
Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☒ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

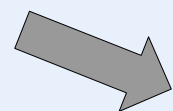


No annotation required (you have it already)

Targeted, or Untargeted with pre-annotated results



Level 3 annotation by (m/z) and retention time



Level 2 annotation by MS/MS + m/z and retention time

Untargeted metabolomics



Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

1) Open:
Metabolomics/OUTPUT/TEST_PTB_vs_HC



Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PT...	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PT...	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PT...	29/06/2026 19:05	File TSV	1 KB
TEST_PT...	29/06/2026 19:05	File TSV	1 KB
TEST_PT...	29/06/2026 19:05	File TSV	1 KB
TEST_PT...	29/06/2026 19:05	File TSV	0 KB
TEST_PT...	29/06/2026 19:05	File TSV	27 KB
TEST_PT...	29/06/2026 19:05	File TSV	49 KB

Only significant
negative hit

Only significant
positive hit

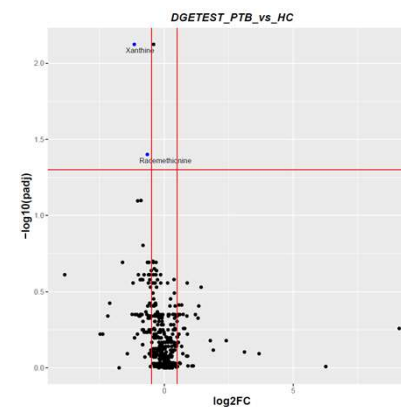
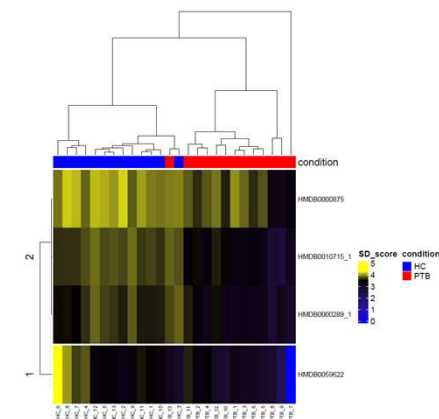
Results table

	A	B	C	D	E	F	G	H	I
1	shapiro_p...	p_val	log2FC	NAME	padj	Name	chemical...	average_m	kegg_id
2	0.082004	2.67E-05	-1.16687	HMDB000	0.007518	Xanthine	C5H4N4O	152.1109	C00385
3	0.778672	0.000297	-0.66172	HMDB003	0.039712	Racemeth	C5H11NO	149.211	C01733



Metabolomics analysis PTB

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

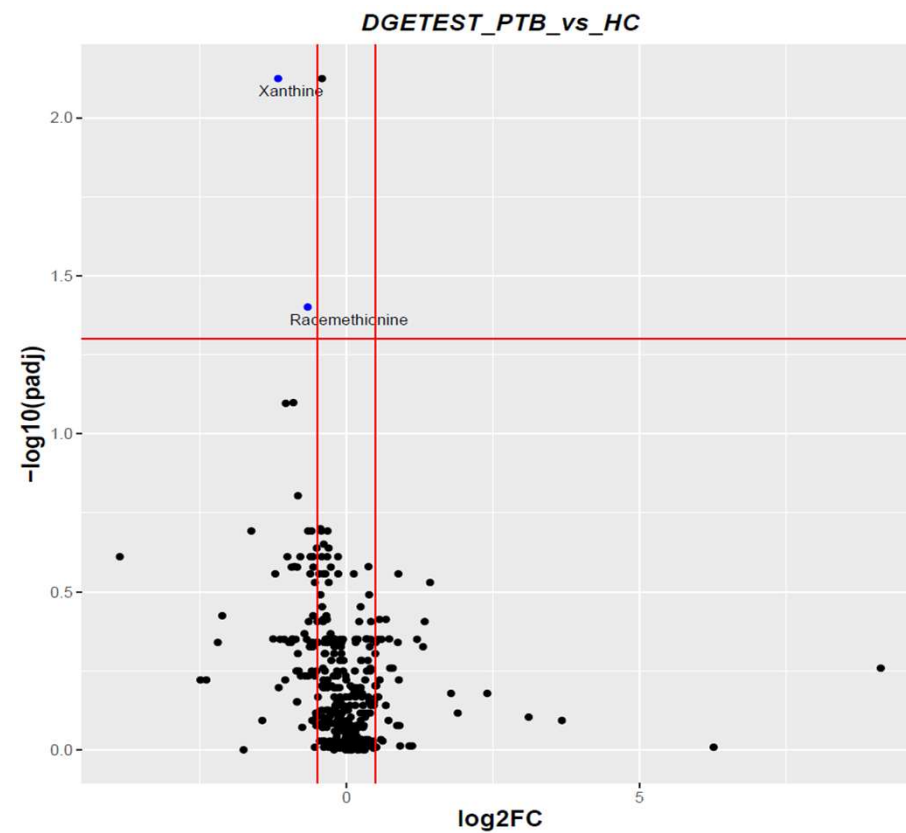




Metabolomics analysis PTB

Nombre

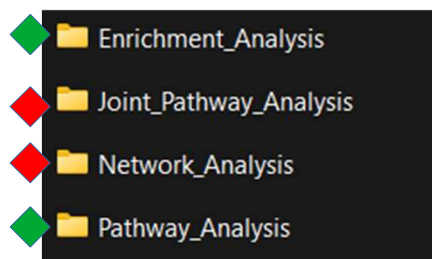
- MetaboAnalyst
- Pathway_analysis
- plot_DMA_PTB_HC_TEST3
- PTB_HC_TEST3_metadata.tsv
- TEST3_PTB_HC_DOWN.tsv
- TEST3_PTB_HC_METABDOWN.tsv
- TEST3_PTB_HC_METABUP.tsv
- TEST3_PTB_HC_UP.tsv
- TEST3_PTB_vs_HC_peak_statistics.tsv
- TEST3_PTB_vs_HC_results.tsv





- ◆ Combined with RNA output
- ◆ Only metabolomics output

Metabolomics analysis PTB



MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB



MetaboAnalyst 6.0

<https://www.metaboanalyst.ca/>



* To avoid to run it again with metabolomics include the transcriptomics input for metaboanalyst in the BiomiX 2.5

Example usage on Metaboanalyst

 **MetaboAnalyst 6.0** - from raw spectra to biomarkers, patterns, functions and systems biology

Home
Data Formats
Tutorials
User Forum
MetaboAnalystR
Publications
Update History
Databases
User Stats
About
Contact



Input Data Type	Available Modules (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		

Enrichment_Analysis
Joint_Pathway_Analysis
Network_Analysis
Pathway_Analysis



* To avoid to run it again with metabolomics include the transcriptomics input for metaboanalyst in the BiomiX 2.5

Example usage on Metaboanalyst



MetaboAnalyst 6.0 - from raw spectra to biomarkers, patterns, functions and systems biology

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[Data Formats](#)
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[User Forum](#)
[MetaboAnalystR](#)
[Publications](#)
[Update History](#)
[Databases](#)
[User Stats](#)
[Data Policy](#)
[About](#)
[Contact](#)


Input Data Type	Available Modules (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features (metabolite list or table)		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		



- Enrichment_Analysis
- Joint_Pathway_Analysis
- Network_Analysis
- Pathway_Analysis

- Transcriptomes_availables
- HMDB_ID_TEST_PTB_vs_HC.tsv

Please upload your data using one of the options below

[Compound List](#) [Concentration Table](#) [Metabolomics Workbench Data](#) [Metabolon Data Sheet](#) [Joint Pathway Analysis](#)

This module is designed for integrative analysis of transcriptomics/proteomics and metabolomics data at pathway level.

- **Metabolomics:**
 - Targeted metabolomics: a list of significant metabolites (or compounds of interest) with optional fold change values;
 - Untargeted metabolomics: a complete peak list. Peak names must be their numeric mass (m/z) values with optional retention times;
- **Transcriptomics / Proteomics:**
 - A list of significant genes (or proteins) with optional fold change values;

Metabolomics Type:

Targeted (compound list) ▼

Compound list with optional fold changes

Copy paste

ID Type:

--- Not Specified --- ▼

Copy paste

Genes/proteins with optional fold changes

ID Type:

--- Not Specified --- ▼

> Submit

[Try our example](#)



Please upload your data using one of the options below

[Compound List](#) [Concentration Table](#) [Metabolomics Workbench Data](#) [Metabolon Data Sheet](#) [Joint Pathway Analysis](#)

This module is designed for integrative analysis of transcriptomics/proteomics and metabolomics data [at pathway level](#).

- **Metabolomics:**
 - Targeted metabolomics: a list of significant metabolites (or compounds of interest) with optional fold change values;
 - Untargeted metabolomics: a [complete](#) peak list. Peak names must be their numeric mass (m/z) values with optional retention times;
- **Transcriptomics / Proteomics:**
 - A list of significant genes (or proteins) with optional fold change values;

Metabolomics Type:

Targeted (compound list) ▼

Compound list with optional fold changes

HMDB0000292	-1.166874014
HMDB0000159	-0.418125595
HMDB0033951	-0.66171578
HMDB0000673_5	-0.906404543
HMDB0000925	-1.035921633
HMDB0001999	-0.82879636
HMDB0011618	-0.444430066
HMDB0000929	-0.595395106
HMDB0010379	-1.623878436
HMDB0010381	-0.429302103
HMDB0008493	-0.322414705
HMDB0008138_11	-0.656334768
HMDB0000738	-0.388022968
HMDB0000267_2	-0.30533277
HMDB0000875	-0.507805485
HMDB0001921	-0.614559907
HMDB0001047	-0.85043600

ID Type:

HMDB ID ▼

Genes/proteins with optional fold changes

ZNF683	-2.302013132
CD69	-1.280647592
CTSW	-1.450083148
CCL4	-1.404571722
ID2	-0.719208523
FEZ1	-1.69089457
NKG7	-1.250631439
HNRNPDL	-0.607121002
FOSB	-3.053290289
PATL2	-0.709080192
GOS2	-2.305585463
CD3D	-0.933529732
NFU1	-0.63829913
XPNPEP2	-1.44088622
GZMM	-1.006252488
SPON2	-1.283466051
GLIS3	-1.330430000

ID Type:

Official Gene Symbol ▼

> Submit

[Try our example](#)



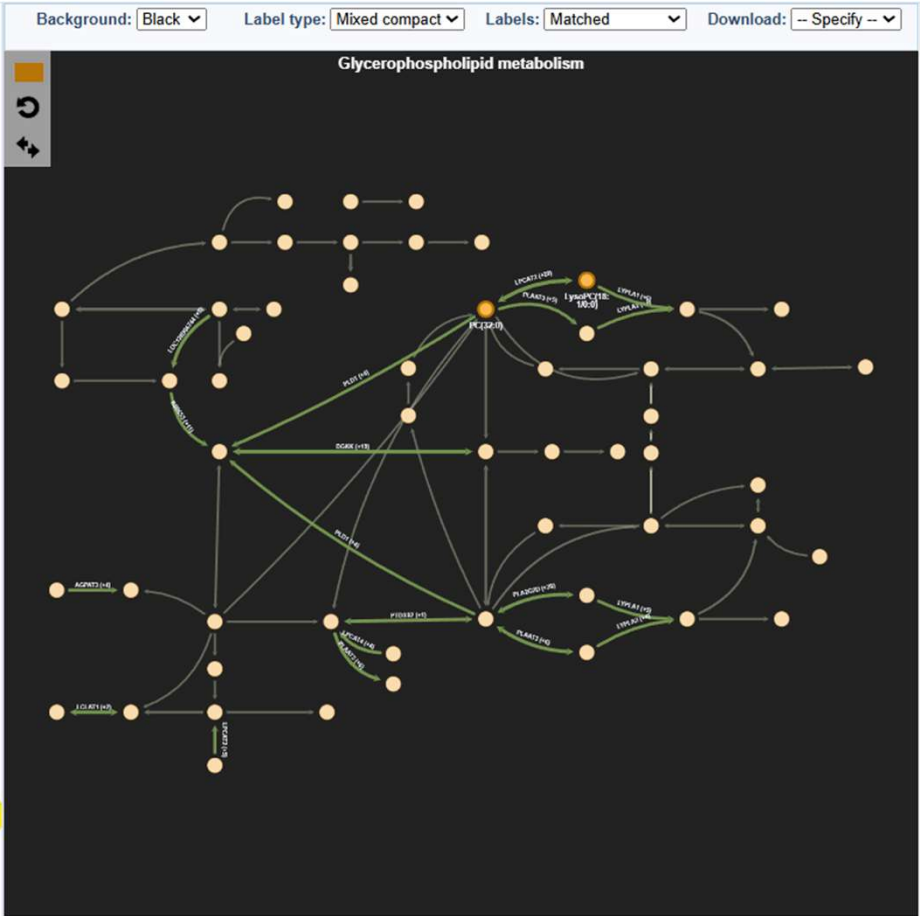
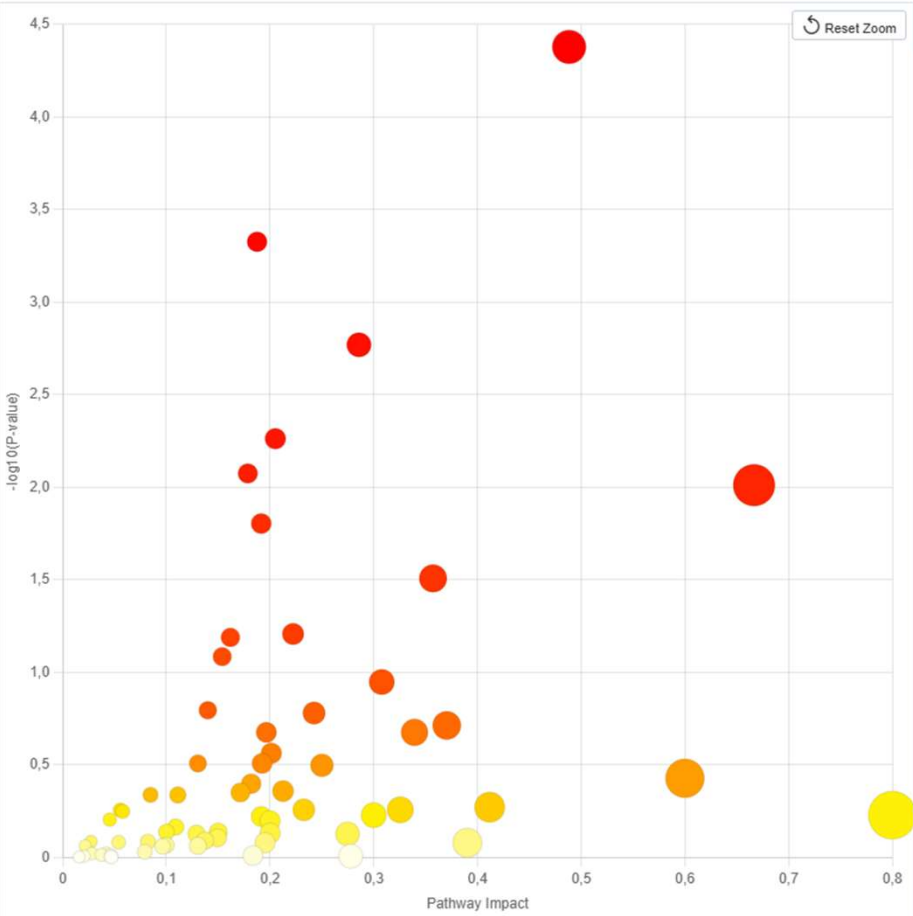


Please upload your data using one of the options below

[Showt Pathway Analysis](#)

Joint Pathway Analysis - Result View:

The scatter diagram summarizes matched pathways by pathway impact (x-axis) and enrichment significance (y-axis, $-\log_{10}$ p-value). Mouse over points to view pathway names and drag to zoom. Use reset to return to default view.



values;
tional retention times;

1 fold changes

002

nbol

> Submit

[Try our example](#)



How to annotate automatically?

2) Load MTBLS7623_positive_mode_unannotated.xls from Metabolomics folder. (Input 1)

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION (•RUN SINGLE OMICS BEFORE)	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Transcriptomics ▼	☐	RNA	☐	Browse file... No file selected
Input 2	☐	☒	Metabolomics ▼	☐	METAB	☐	Browse file... ✔ MTBLS7623_positive_mode_unannotated.tsv /shared/MTBLS7623_positive_mode_unannotated.tsv



How to annotate automatically?

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☒ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

1) In Advance Options select the “MS1 only”

METABOLITE ANNOTATION

Metabolite annotation column

HMDB ▼



Positive or Negative mode?

Which adduct do you expect?

Which databases you want to consult to retrieve the annotation?

Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)

☒ MS1 only (non-targeted m/z matching against databases)

☐ MS1 + MS2 (m/z matching + spectral matching)

MS1 MATCHING PARAMETERS

Ion mode: positive M/Z tolerance (ppm): 15

MS1 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS1 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS1

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 ANNOTATION FILE (OPTIONAL)

Annotation file: Browse... Input index: No

The annotation should be performed on which input position?

Error match tolerance between “your” peak and the database match



Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)

☒ MS1 only (non-targeted m/z matching against databases)

☐ MS1 + MS2 (m/z matching + spectral matching)

MS1 MATCHING PARAMETERS

Ion mode: positive M/Z tolerance (ppm): 15

MS1 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS1 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS1

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 ANNOTATION FILE (OPTIONAL)

Annotation file: Browse... Input index: No

3) Load
MTBLS7623_positive_mode_MS1_annotation.tsv
from Metabolomics folder. (Input 1)

	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258



-> Request of annotation to Ceu
Mass Mediator Server

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
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PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB



Annotation_metabolites_B



TEST_PTB_vs_HC_results.tsv 29/06/2026 19:05 File TSV 49 KB

Adduct type

HMDB, KEGG and
PubChem ID and links

Statistics

Formula

Ppm error:
How much reliable?

Annotated name

m/z
Peak
number

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	m/z	NAME.x	shapiro_p_val		log2FC	RT_sec	RT_min	padj	retention	adduct_re	RT_score	identifier	name	formula	adduct	molecular	error_ppm	ionization	final_scor	KEGG	KEGG_UR	HMDB	HMDB_UR	LipidMaps	LipidMaps	Metlin
2	320.959	peak_114	3.614309	0.00013	#NOME?	60.265	1.00442	0.0674	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	117280	3-[3,5-dih	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
4	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	58543	3-hydroxy	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
5	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	69877	5-[3,5-dih	C14H10O	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			



How to annotate automatically using fragmentation data (MS/MS)?

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☒ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

1) In Advance Options select the “MS1 + MS2”

METABOLITE ANNOTATION

Metabolite annotation column

HMDB ▼



Error match tolerance between
“your” peak and the database
match

Error match tolerance between “your”
peak m/z in MS1 and in MS2 data

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

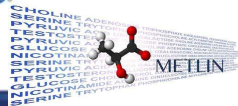
MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



Public database priority in the MS2 match
(available: HMDB, MONA, Massbank)



Public database priority in the MS1 match
(available: HMDB, KEGG, Lipidmaps, Metlin)



Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablob SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

- ☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

- ☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

- ☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablob SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

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HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...

3) Load
MTBLS7623_positive_mode_MS1_annotation.tsv
from Metabolomics folder. (Input 1)

	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258



MS2_PLASMA1_1.mzML

MS2_URINE2_1.mzML

MS2_PLASMA_pool_MSMS-r001.mgf

pool_MSMS-r001.mgf

mgf_files

mzML_files

-> Provide the directory of the
MzML file or mgf file(s)

Name these file(s) including the
label used in the "label" i.e (TEST,
or PLASMA)

Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablob SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

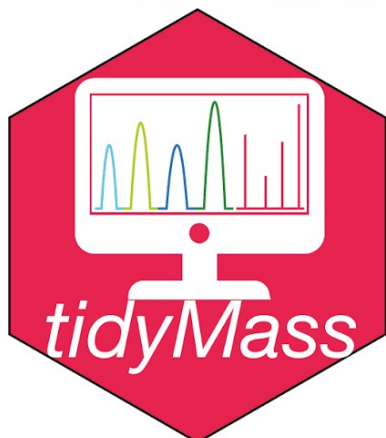
HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



-> Request of annotation to Ceu
Mass Mediator Server

-> Local MS2 spectra extraction
and match on the databases
selected (locally available).

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

+ Annotation_metabolites_B

+ MS2_SPECTRA

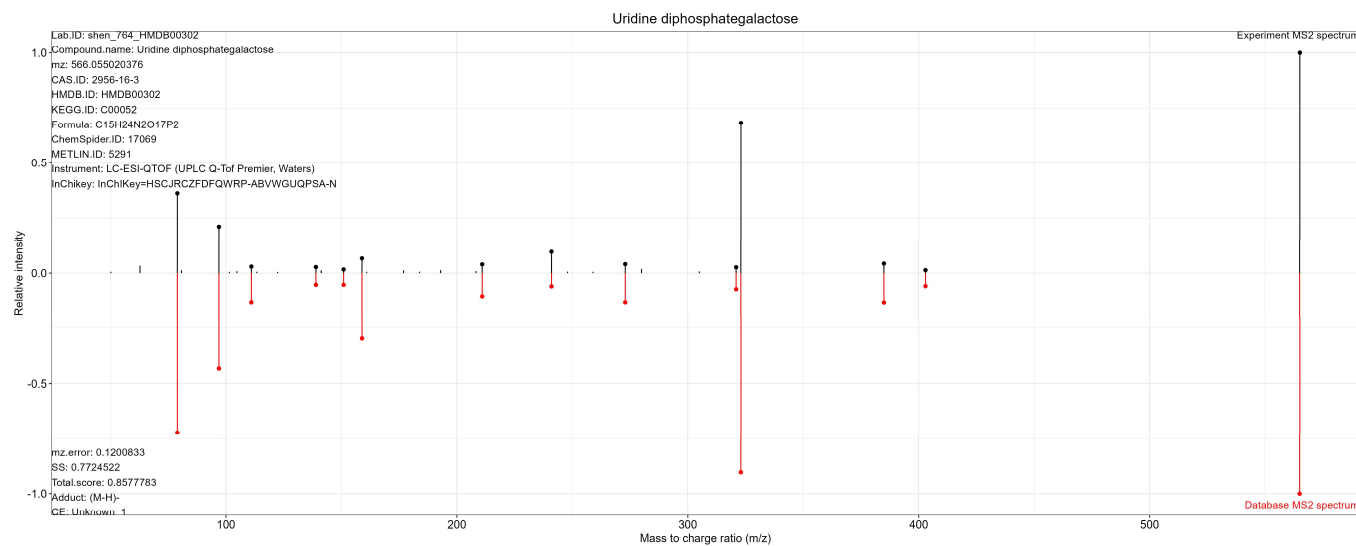


MS2_SPECTRA

ms2_match_plot_HMDB

ms2_match_plot_MASSBANK

ms2_match_plot_MONA



-> Results updated with MS2 annotation and folder including spectral match of the annotated peaks

HMDB_URI	LipidMaps	LipidMaps	Metlin	Metlin_URI	PubChem	PubChem	MS2_annot	annot_level
/dbget-bin/www_bget?cpd:C027			3596	https://me	11988267	https://pub	NA	Level_3
/dbget-bin/www_bget?cpd:C018			65671	https://me	25200469	https://pub	NA	Level_3
http://www.hmdb.ca/metabolite:			62907	https://me	439795	https://pub	NA	Level_3
http://www.hmdb.ca/metabolite:			3595	https://me	192	https://pub	NA	Level_3

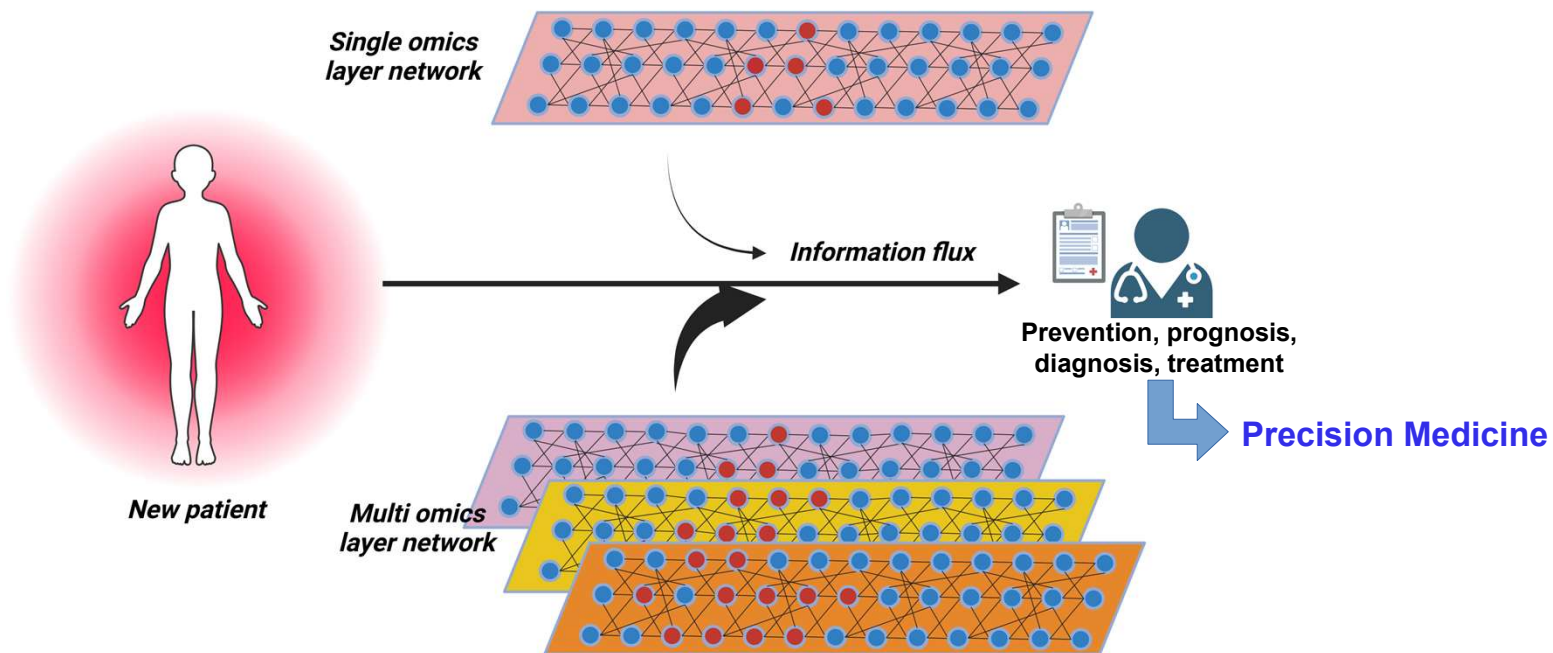


Part 2 – Multi-Omics Integration





Multi-omics data integration, the grail of biology understanding ?



Pros

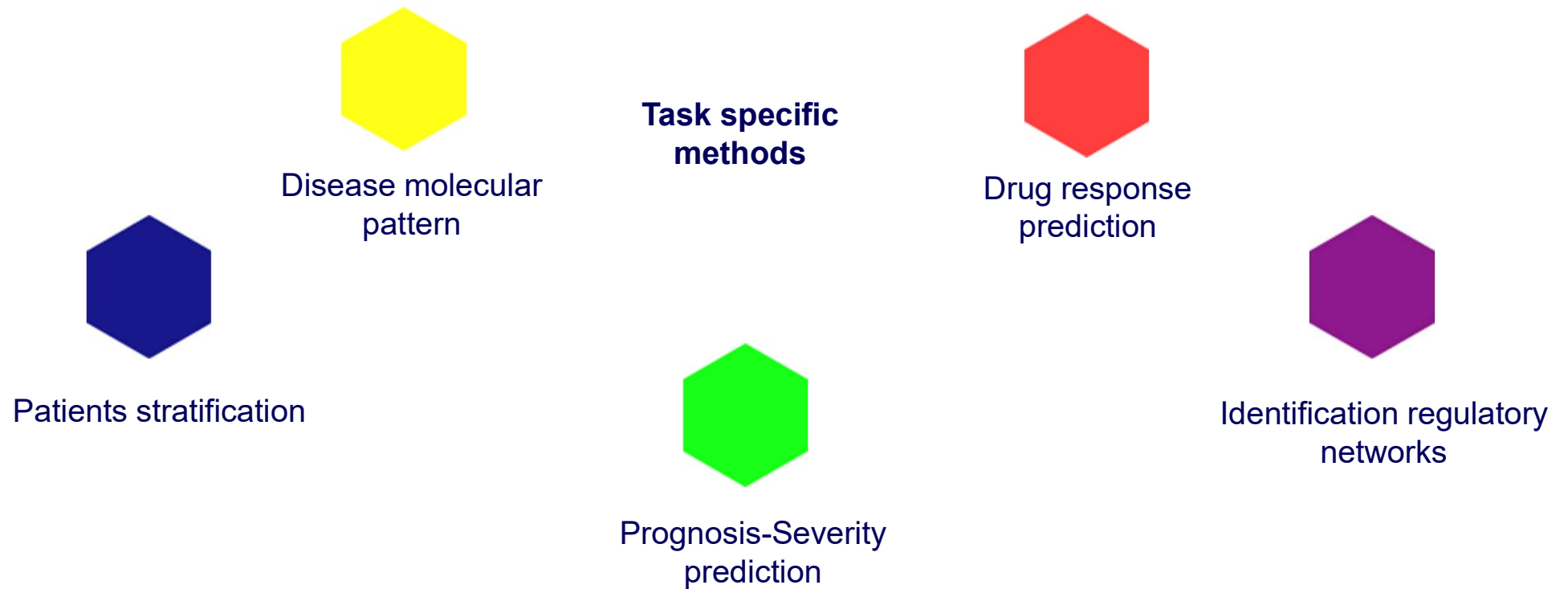
- Observations strengthened
- Holistic interpretation
- Detailed characterization

Cons

- Different distribution
- Curse of dimensionality
- Missing data

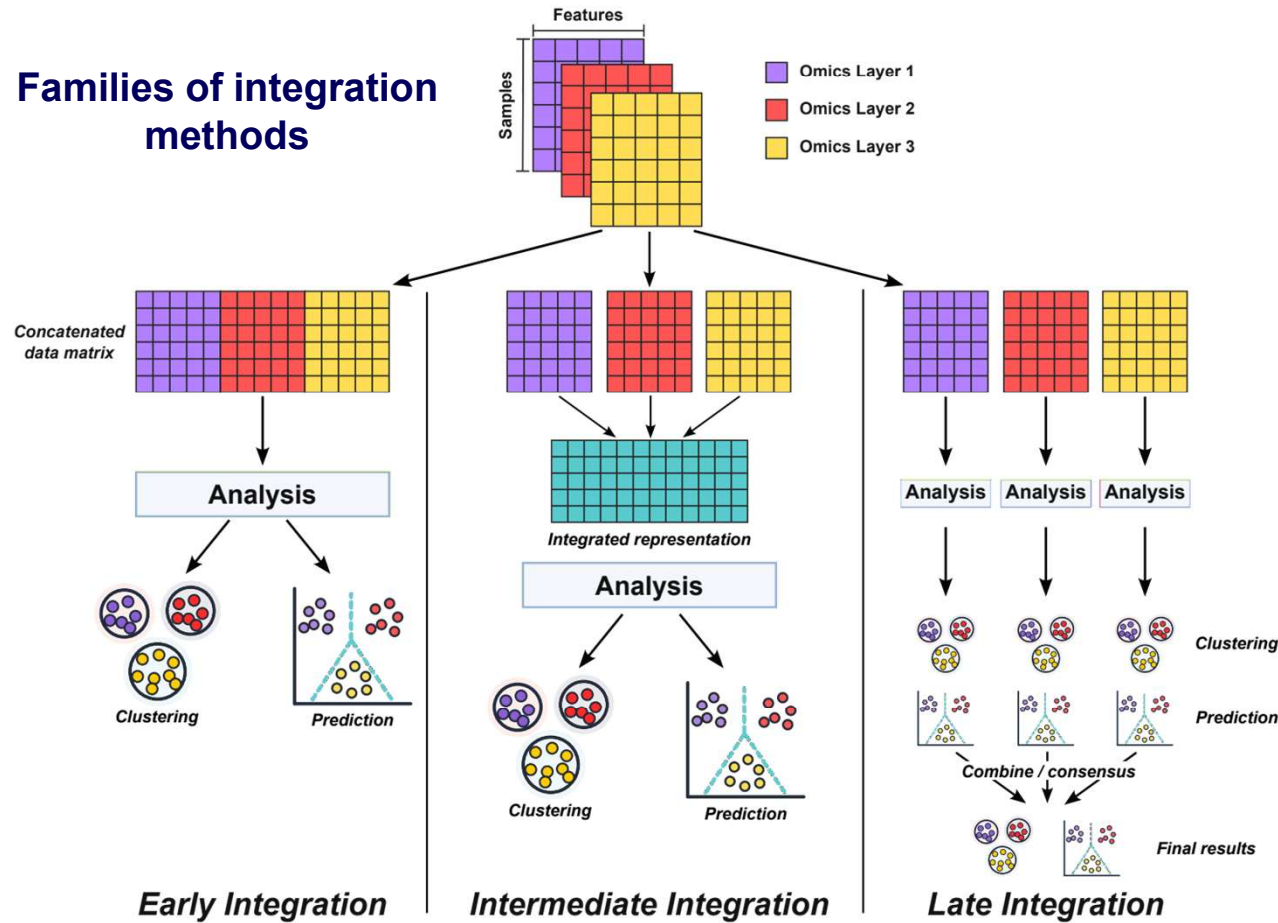


Multi-omics data integration, the grail of biology understanding?





Families of integration methods

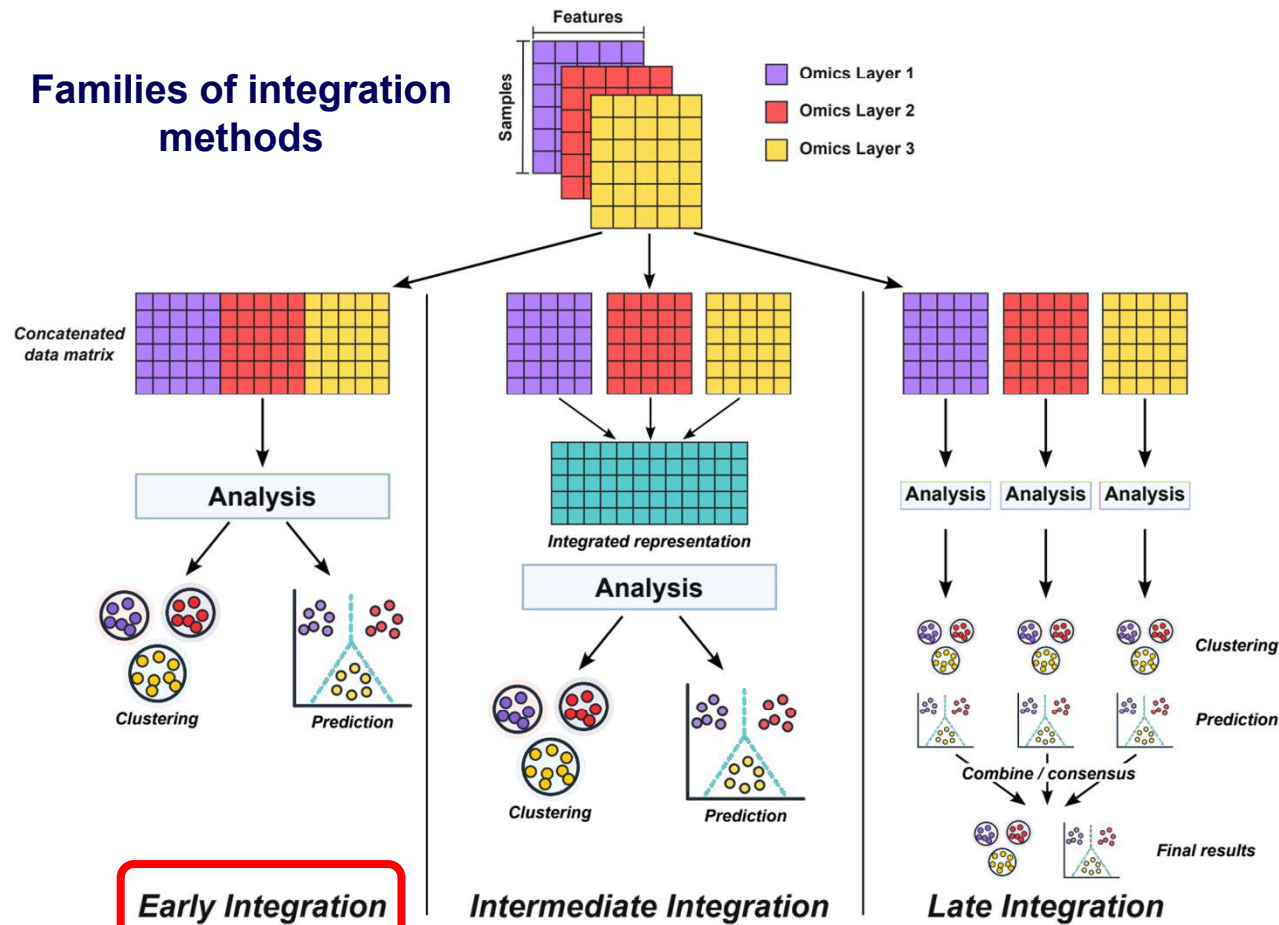


Key distinction is the **stage** at which information from different omics is integrated:

Ritchie MD et al., Nat Rev Genet, 2015; Zitnik M et al., Inf Fusion, 2019



Families of integration methods



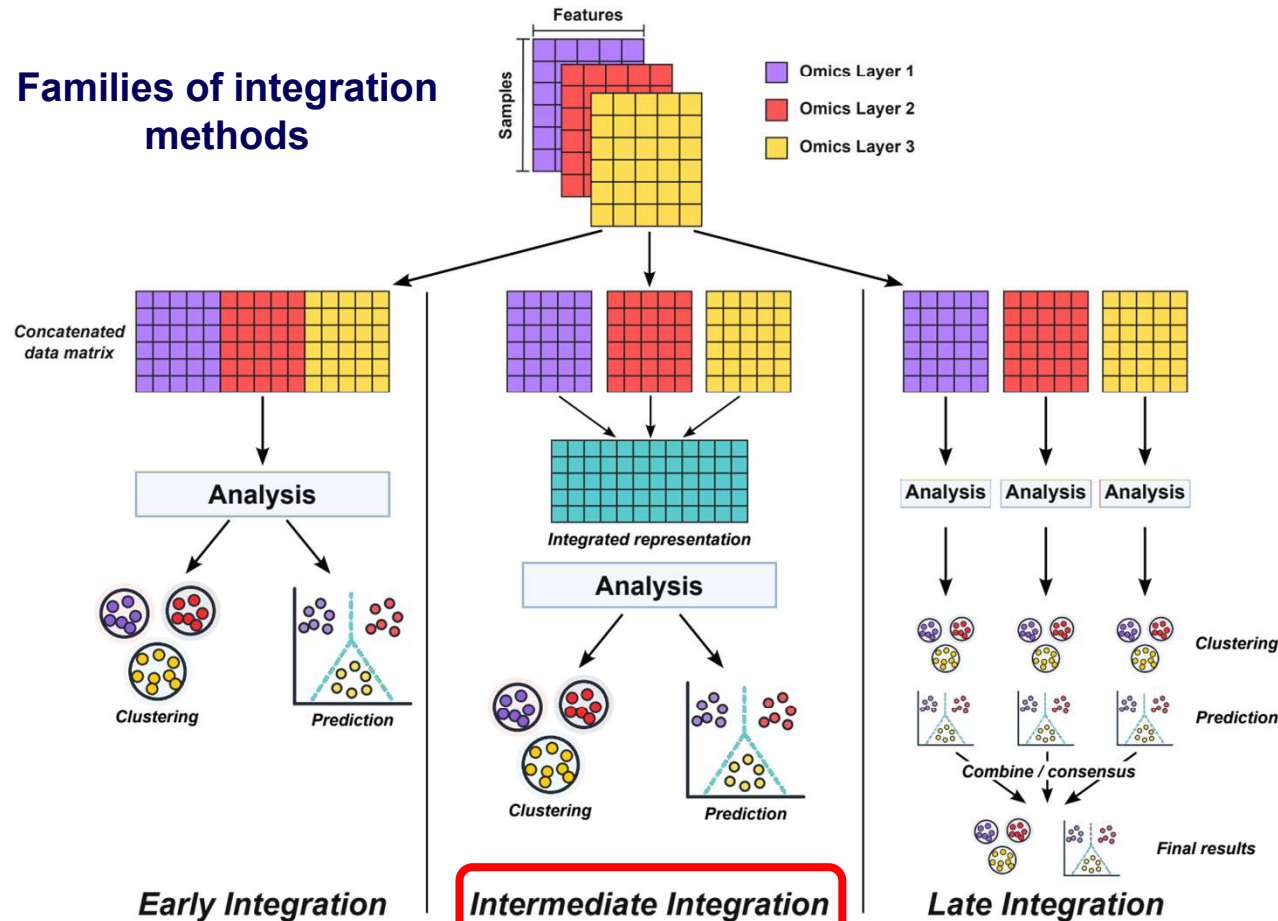
Early integration:

- Different omics measured on the same samples
- Features from the omics are concatenated
- A joint downstream analysis is performed on the integrated matrix

Integration occurs at the input level



Families of integration methods



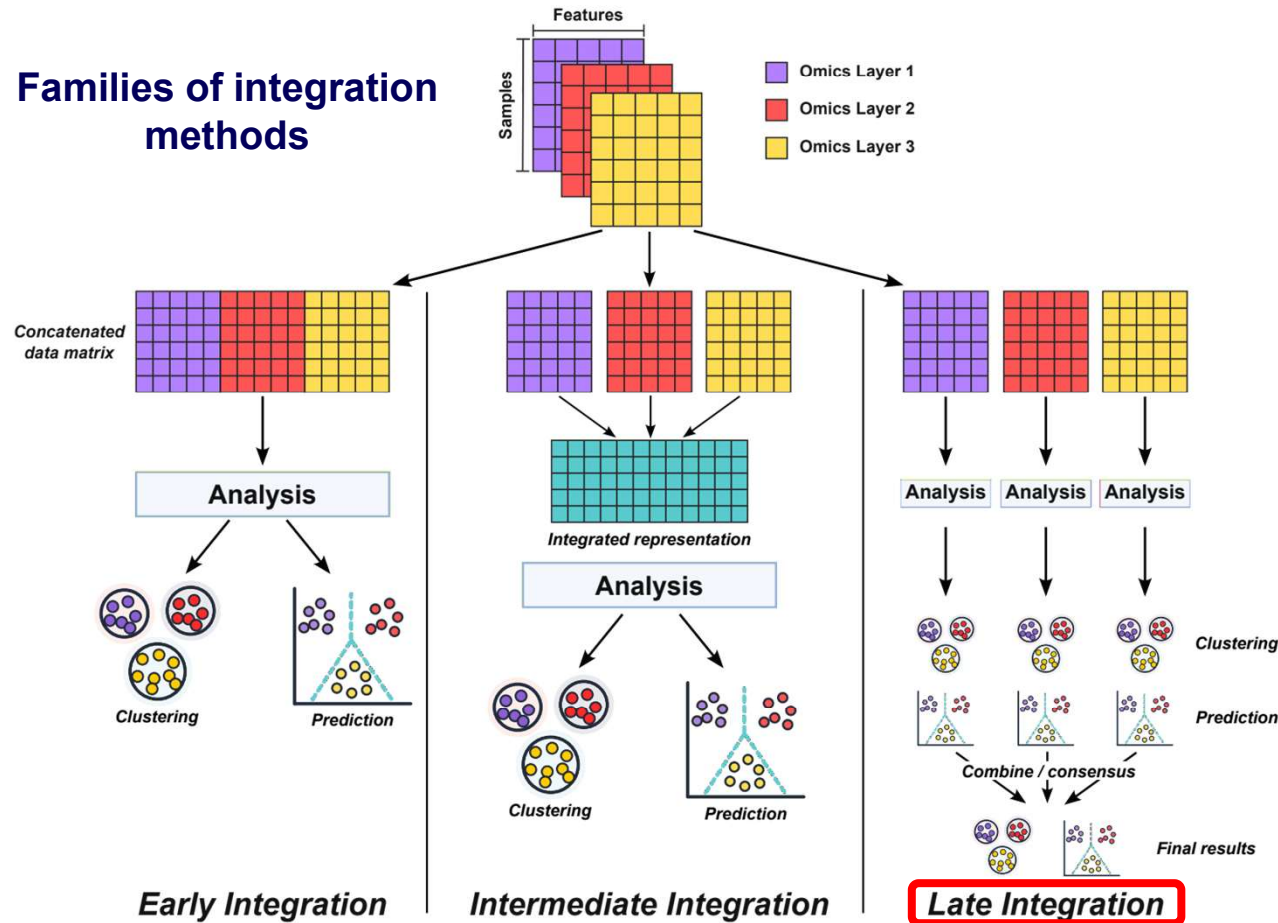
Intermediate integration:

- Omics are retained as separate data layers
- Information across layers is integrated while learning or constructing a joint representation
- The resulting integrated representation is used for downstream analysis

Integration occurs at the representation level



Families of integration methods



Late integration:

- Each omic layer is analyzed separately
- The separate outputs are then combined and used to derive a consensus

Integration occurs at the output level



Methods can be classified based on the employed **computational approach**.

In BiomiX, the following intermediate approaches are implemented:

- **MOFA** (Multi-Omics Factor Analysis)
- **DIABLO** (Data Integration Analysis for Biomarker discovery using Latent cOmponents)
- **SNF** (Similarity Network Fusions)
- **NEMO** (NEighborhood based Multi-Omics clustering)

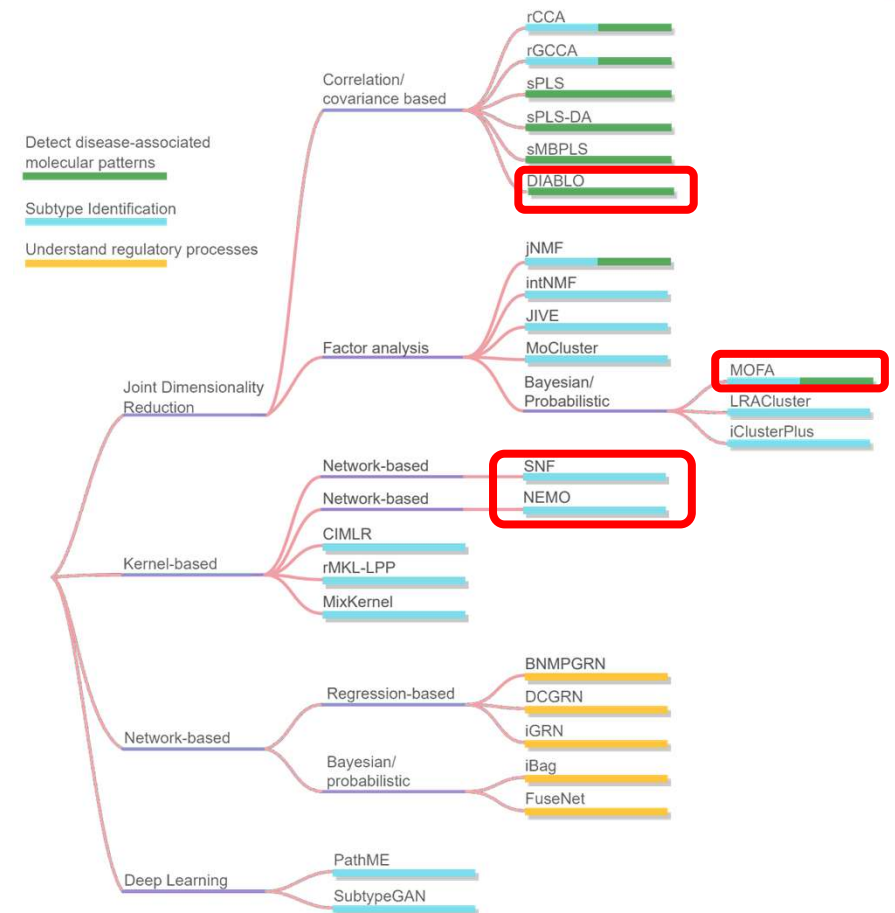


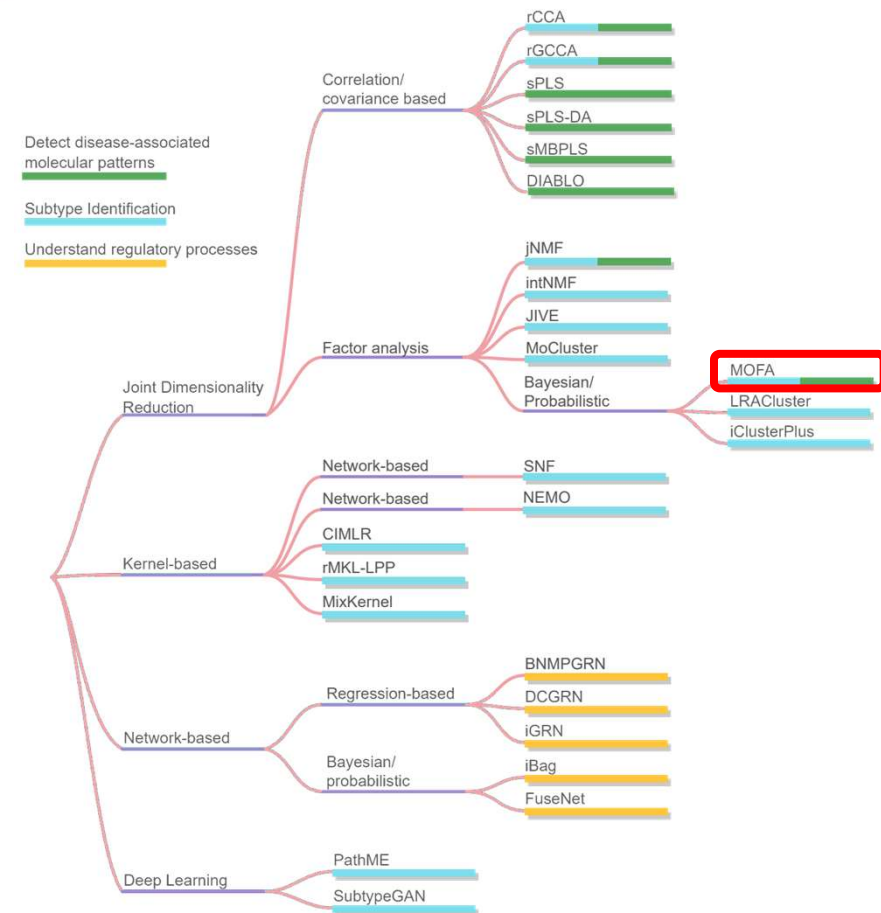
Image adapted from Athieniti E, Spyrou GM, Comput. Struct. Biotechnol. J, 2023



MOFA (Argelaguet R, et al. Mol. Syst. Biol., 2018)

MOFA is an **unsupervised Bayesian factor-analysis framework** that learns a low-dimensional representation of multi-omics data through latent factors **capturing shared and modality-specific sources of variation**.

- Data **integration suited for** Gaussian, Bernoulli and Poisson **distributions**
- **Missing omics** data handled
- **Easy MOFA factor interpretation** enabled by sparse weights
- Low performance on strongly non-linear data
- Factor selection depends on user-defined training criteria





Multiple omics matrices Y^m are approximated using a **common sample-factor matrix Z** , but each omic has its own **feature-weight matrix $W^{(m)}$** .

$$Y^{(m)} \approx ZW^{(m)T} \quad m = 1, \dots, M$$

$Y^{(m)}$: one data matrix for each omic m

Z : one common factor matrix for the samples

$W^{(m)}$: one loading/weight matrix for each omic m

Factor-wise and feature-wise **sparsity** lead to more interpretable solutions!

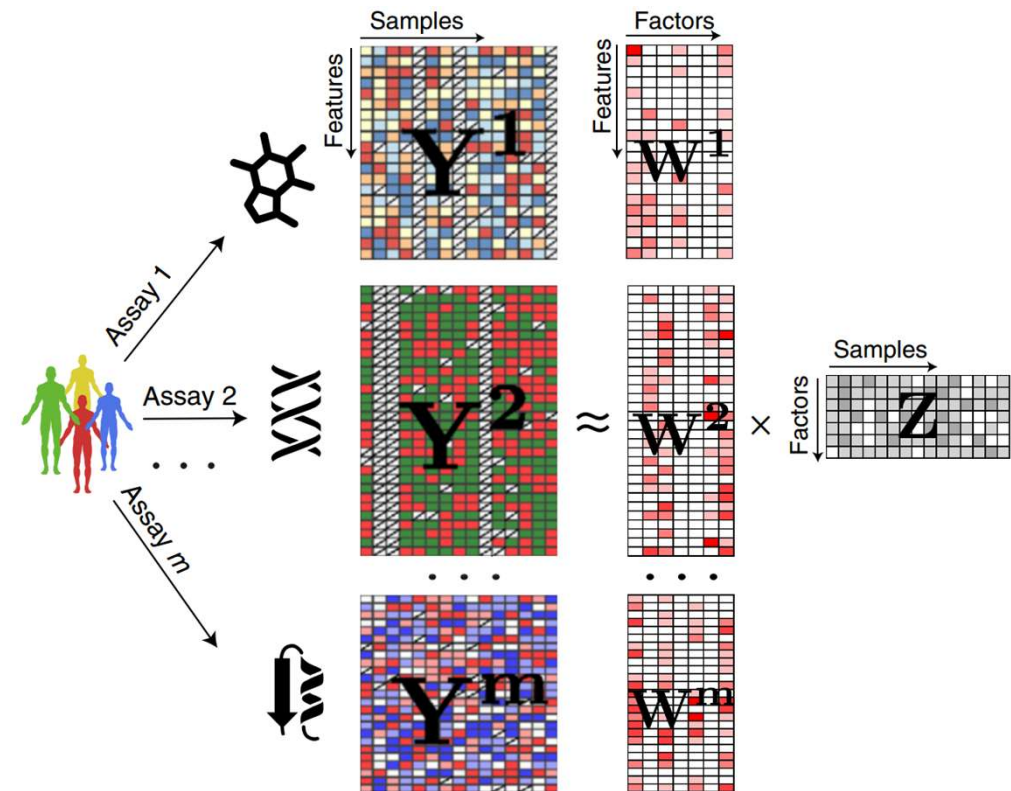


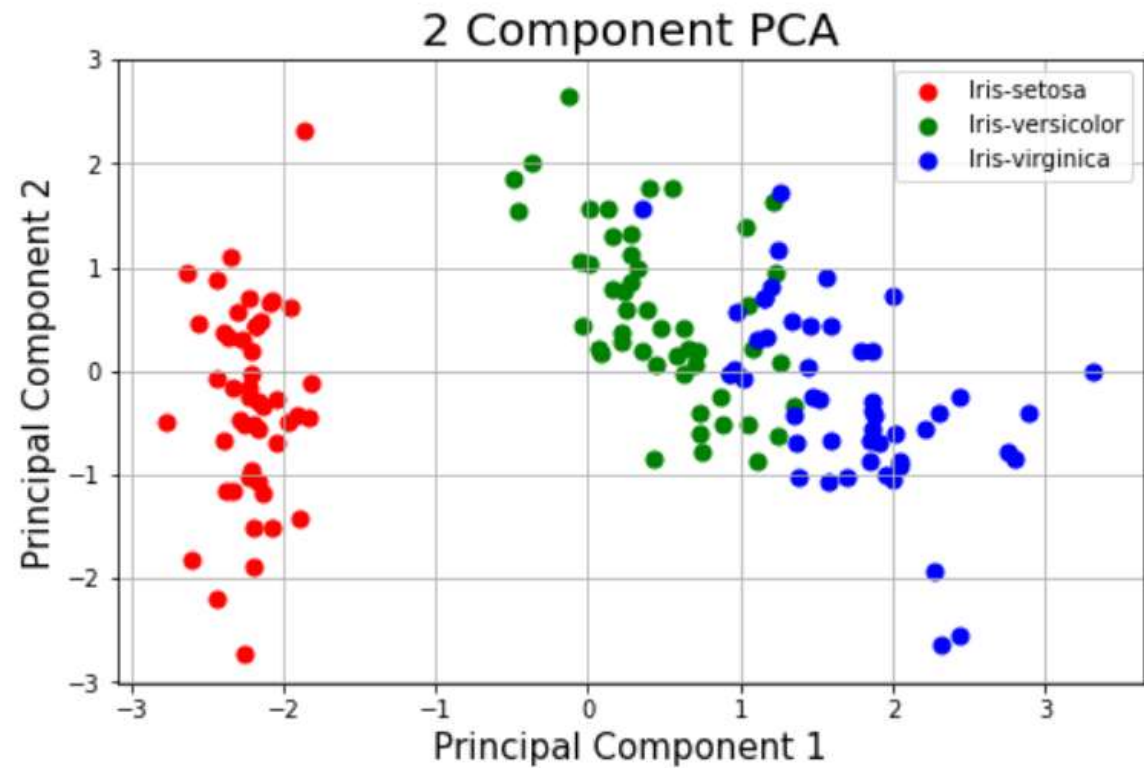
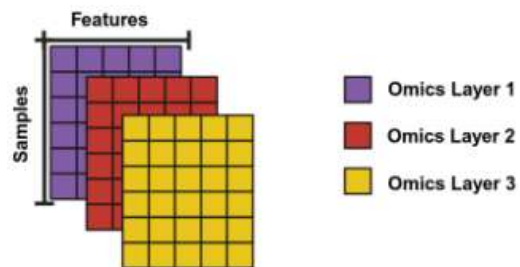
Image from Argelaguet R, et al. Mol. Syst. Biol. 2018



To make it simple

- MOFA is a dimensional reduction like PCA !

But more complicated
(bayesian model) and
including multiple omics

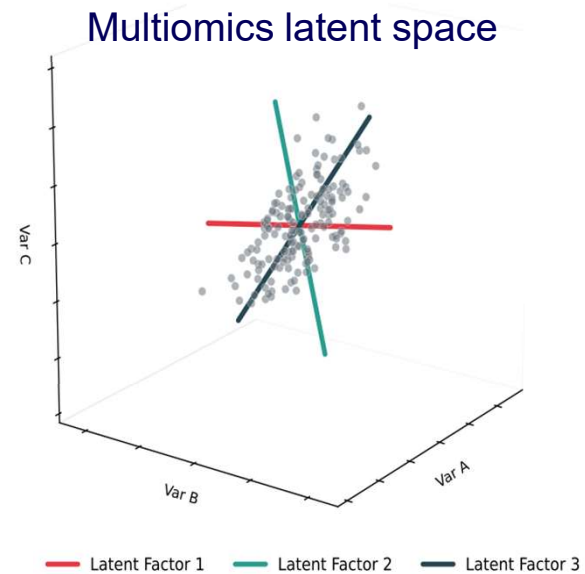




To make it simple

- MOFA is a dimensional reduction like PCA !

But more complicated
(bayesian model) and
including multiple omics



MOFA select the factors
with balanced weight from
all the omics.



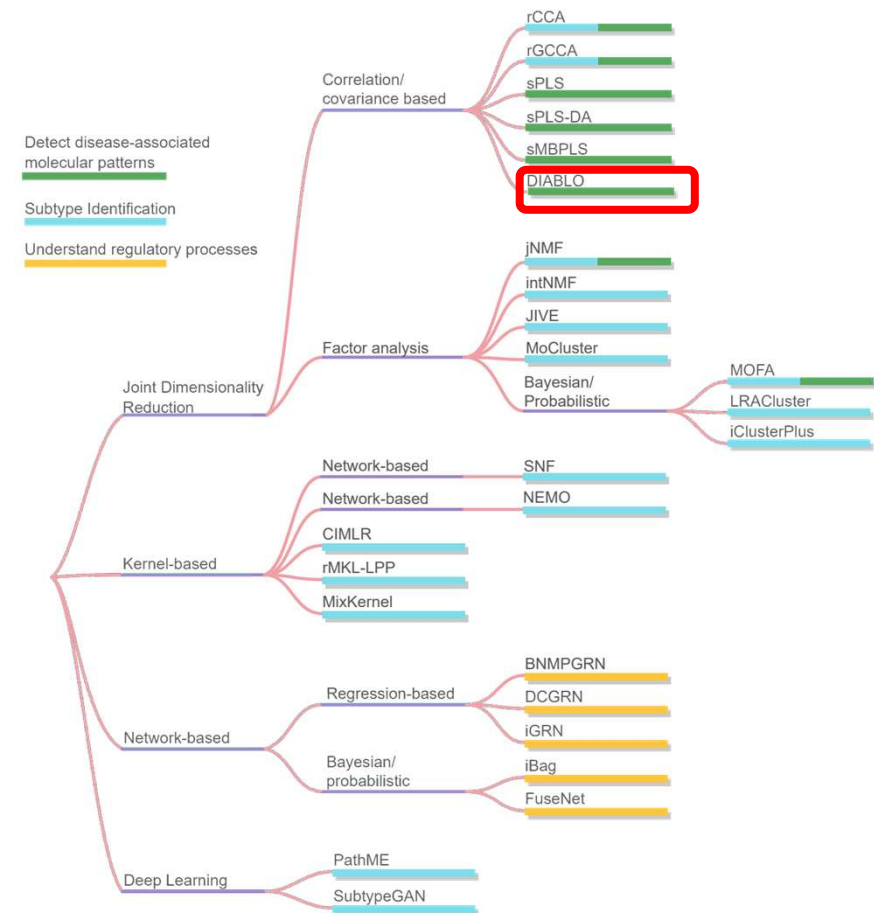
DIABLO (Singh A, et al. Bioinformatics, 2019)

DIABLO is a supervised extension of sGCCA (sparse Generalized Canonical Correlation Analysis).

Objectives:

1) **maximize the covariance** between linear combinations of variables (latent components) & **maximize groups discrimination**

- Joint integration and supervised feature selection (LASSO-like)
- Identifies correlated multi-omics signatures associated with phenotype
- Low performance on non-linear data
- Training requires matched samples across omics





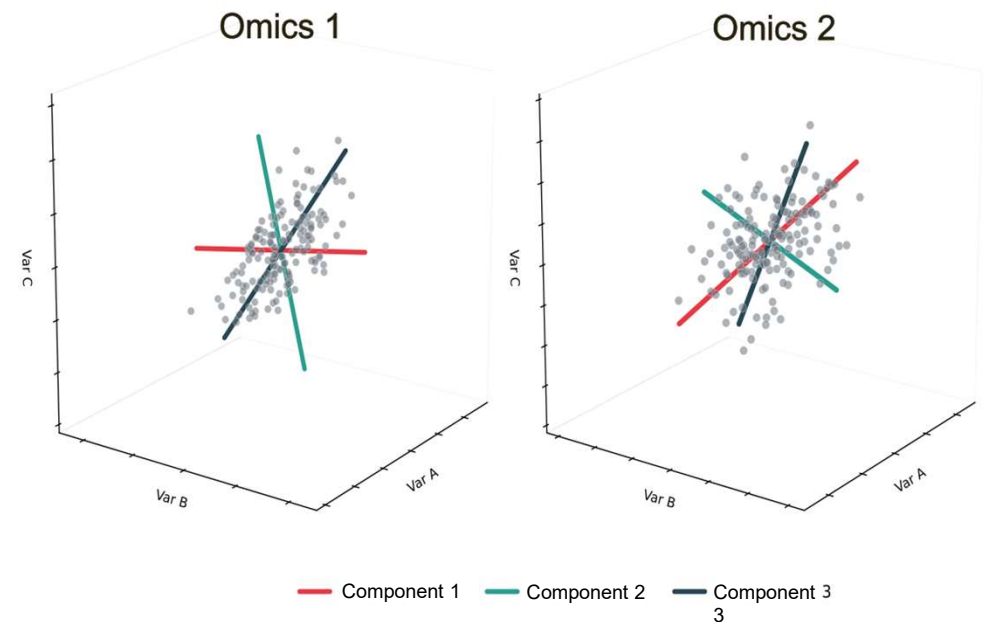
- For each omic, DIABLO learns a latent component as a **weighted combination of features**.

For sample i and omic q , we obtain the component score $t_i^{(q)}$ as:

$$t_i^{(q)} = a_1^{(q)}x_{i1}^{(q)} + a_2^{(q)}x_{i2}^{(q)} + \dots + a_p^{(q)}x_{ip}^{(q)}$$

- Sparsity is imposed on the weights**, so many coefficients can become zero (i.e. feature selection)
- Components are **learned jointly** to maximize association across connected omics while discriminating the phenotype.

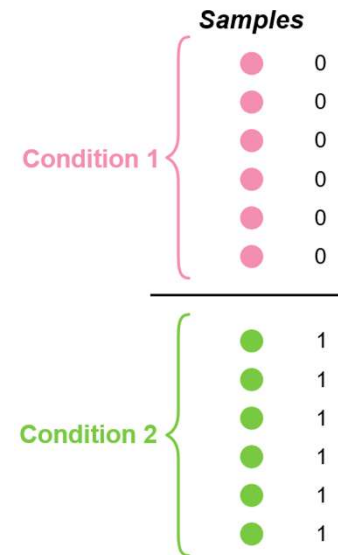
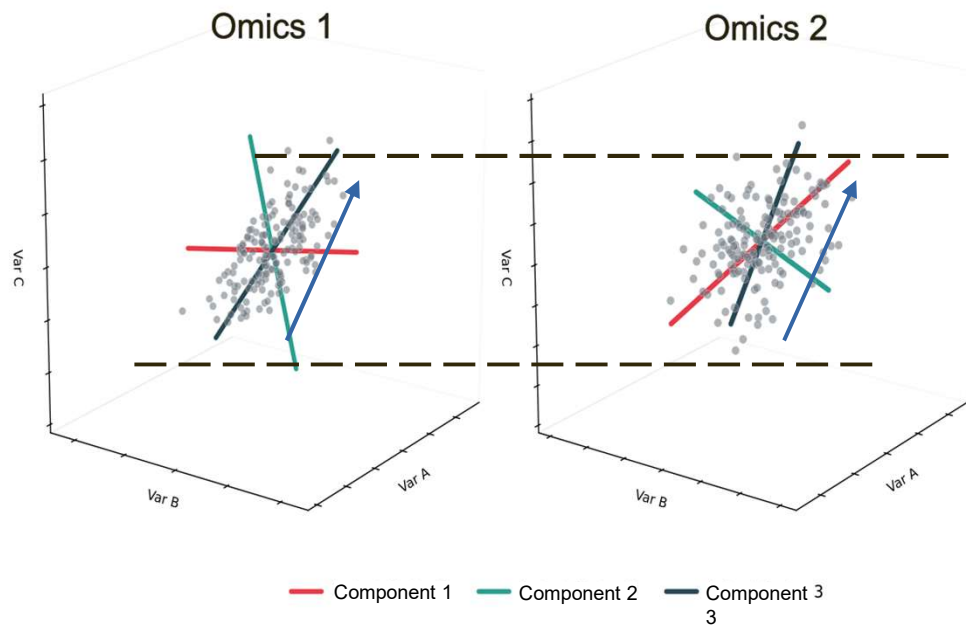
Each omic has its own components!





Dummy matrix → provides the supervised frame to select the components discriminating the two conditions

Design matrix → defines how strongly we expect pairs of omics to be related (high assoc. = 1, low assoc. = 0)



Dummy matrix

Samples	Cond. 1	Cond. 2
S ₁	1	0
S ₂	0	1
...	...	...
S _N	1	0

Design matrix

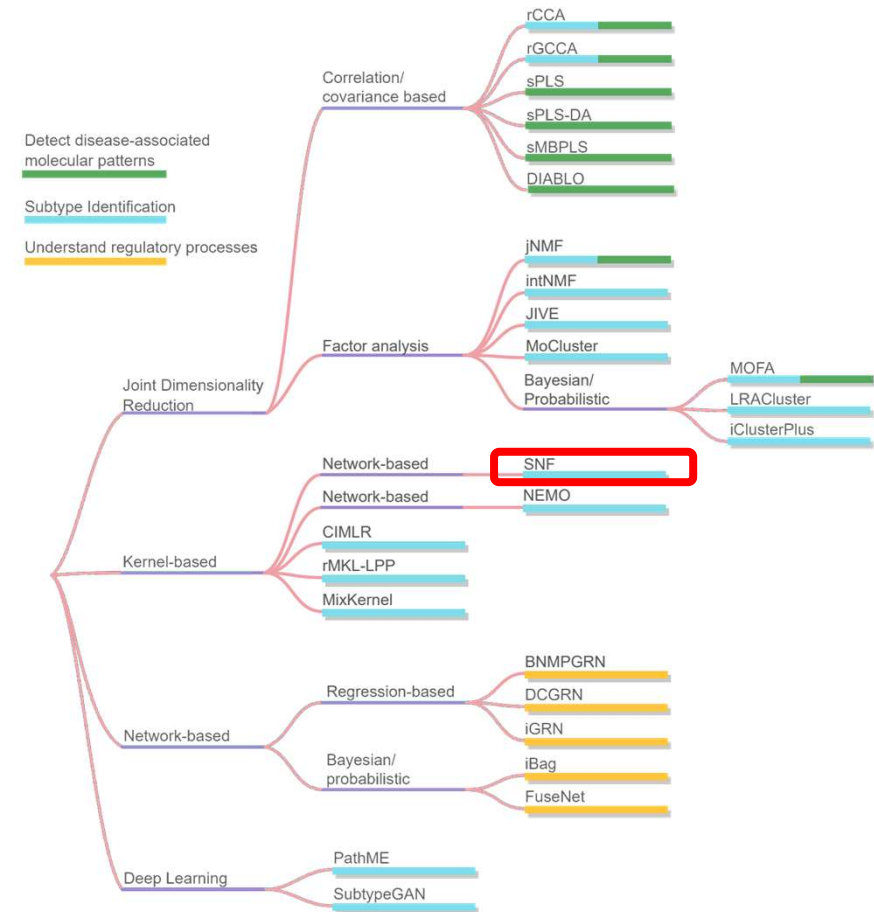
	RNA	Methylome
RNA	0	0.8
Methylome	0.8	0



Similarity Network Fusion (SNF) (Wang B, et al., Nature Methods, 2014)

SNF is an unsupervised multi-omics integration method that builds one sample similarity network per omic and iteratively fuses them into a single **integrated network** for **subtype discovery**, exploiting a message-passing process.

- Unsupervised subtype discovery
- High-dimensional features into an integrated sample-sample network
- Capture local/nonlinear relationships
- Network interpretation
- Limited biomarker interpretation
- Arbitrary choice of hyperparameters
- No missing-data handling
- Poor scaling with number of samples





How SNF works

1) **Omic-specific similarity matrix W** is computed.

2) Computation of **omic-specific global similarity matrix P** :

$$P(i, j) = \begin{cases} \frac{W(i, j)}{2 \sum_{k \neq i} W(i, k)} & , \text{ if } j \neq i \\ 1/2 & , \text{ if } j = i \end{cases}$$

3) Computation of the **omic-specific local similarity matrix S , retaining only k -nearest neighbours**:

$$S(i, j) = \begin{cases} \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)} & , \text{ if } j \in N_i \\ 0 & , \text{ otherwise} \end{cases}$$

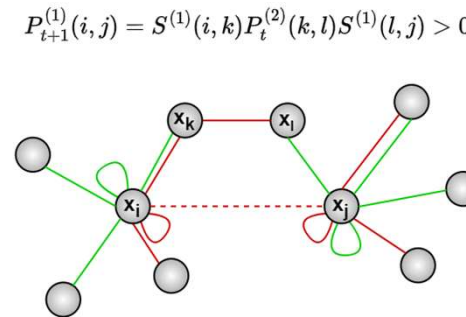
where $N_i = \{x_k | x_k \in kNN(x_i) \cup \{x_i\}\}$.

4) Integration through an **iterative parallel diffusion process and final average** (two views $s=\{1,2\}$):

$$\begin{aligned} P_{t+1}^{(1)} &= S^{(1)} \times P_t^{(2)} \times S^{(1)T} \\ P_{t+1}^{(2)} &= S^{(2)} \times P_t^{(1)} \times S^{(2)T} \end{aligned} \quad \longrightarrow \quad P_{t+1}^{(1)}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(1)}(i, k) P_t^{(2)}(k, l) S^{(1)}(l, j)$$

5) **Spectral clustering** for subgroup discovery.

Case I: indirect connection in $P^{(2)}$



$$P_{t+1}^{(1)}(i, j) = S^{(1)}(i, k) P_t^{(2)}(k, l) S^{(1)}(l, j) > 0$$

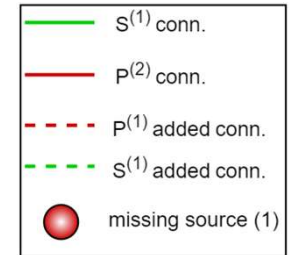


Image adapted from Gliozzo J, et al Bioinformatics, 2025

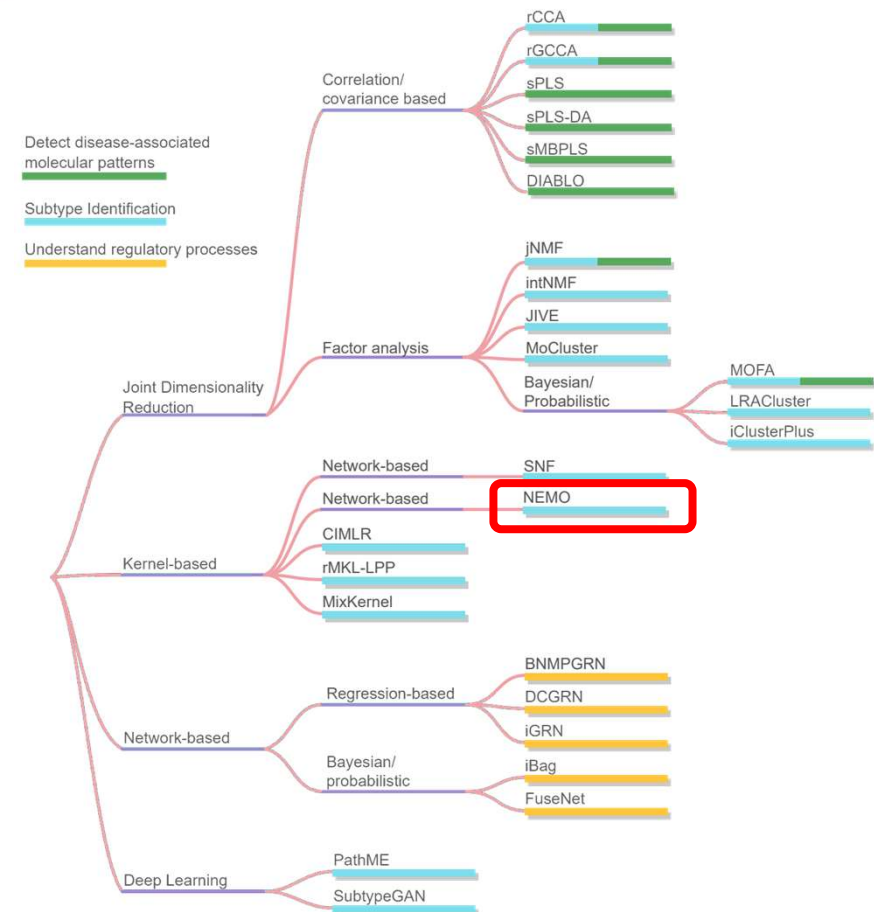


NEighborhood based Multi-Omics clustering (NEMO)

(Rappoport and Shamir, Bioinformatics, 2019)

NEMO is an **unsupervised network-based integration** approach that fuses local sample-neighbourhood similarities across omic-specific networks for **subtype discovery**.

- Unsupervised subtype discovery
- High-dimensional features into an integrated sample-sample network
- Capture local/nonlinear relationships
- Sample-sample network interpretation
- **Handles partial multi-omics datasets**
- Limited biomarker interpretation
- Choice of hyperparameters
- Assumes each pair of samples shares at least one measured omic





How NEMO works

1) Omic-specific similarity matrix using “**scaled exponential Euclidean kernel**” (same as SNF).

2) Define a **relative similarity matrix** for each omic l :

$$RS_l(i, j) = \frac{S_l(i, j)}{\sum_{r \in \eta_{li}} S_l(i, r)} \cdot I(j \in \eta_{li}) + \frac{S_l(i, j)}{\sum_{r \in \eta_{lj}} S_l(r, j)} \cdot I(i \in \eta_{lj})$$

Similarity is evaluated relative to the local neighborhoods of both samples (i,j).

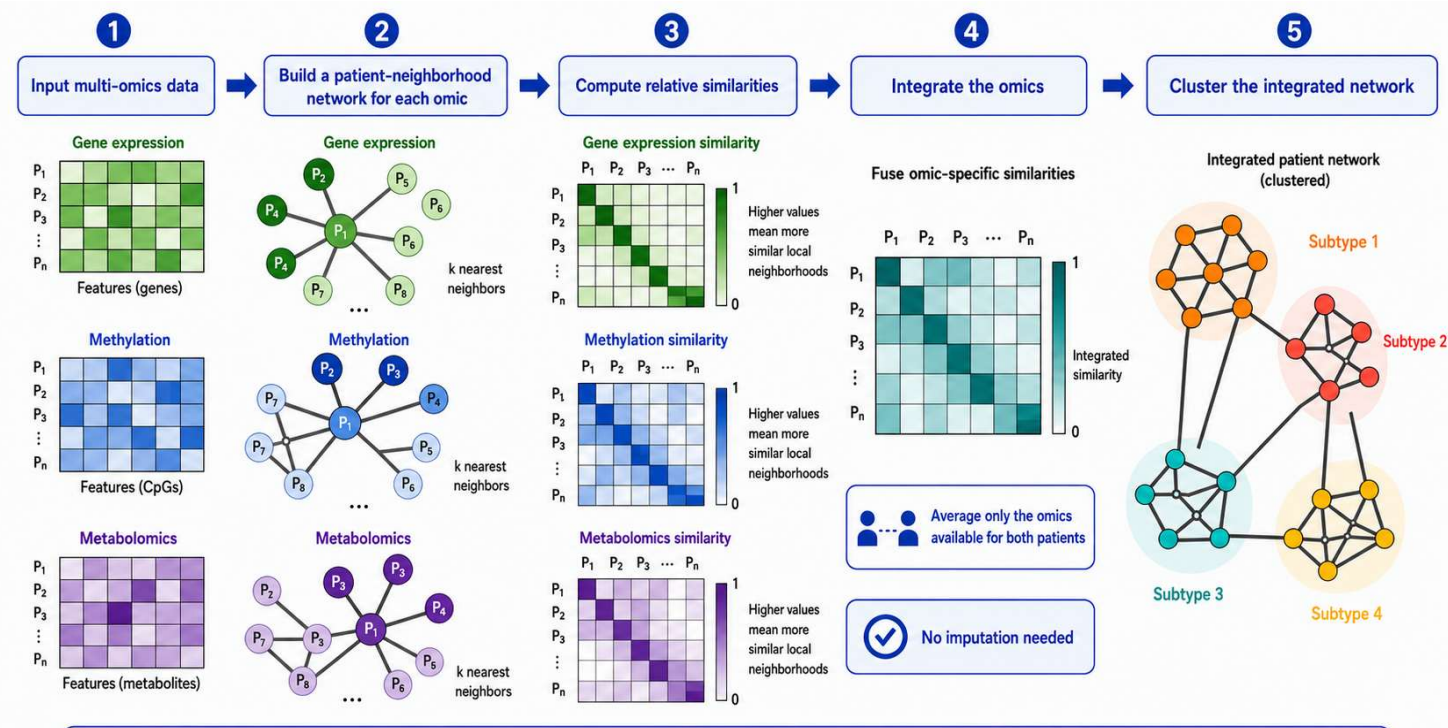
3) Integration by **average** considering **omics types available for both samples JM(i,j)**:

$$ARS(i, j) = \frac{1}{|JM(i, j)|} \sum_{l \in JM(i, j)} RS_l(i, j)$$

4) **Spectral clustering** for subgroup discovery.

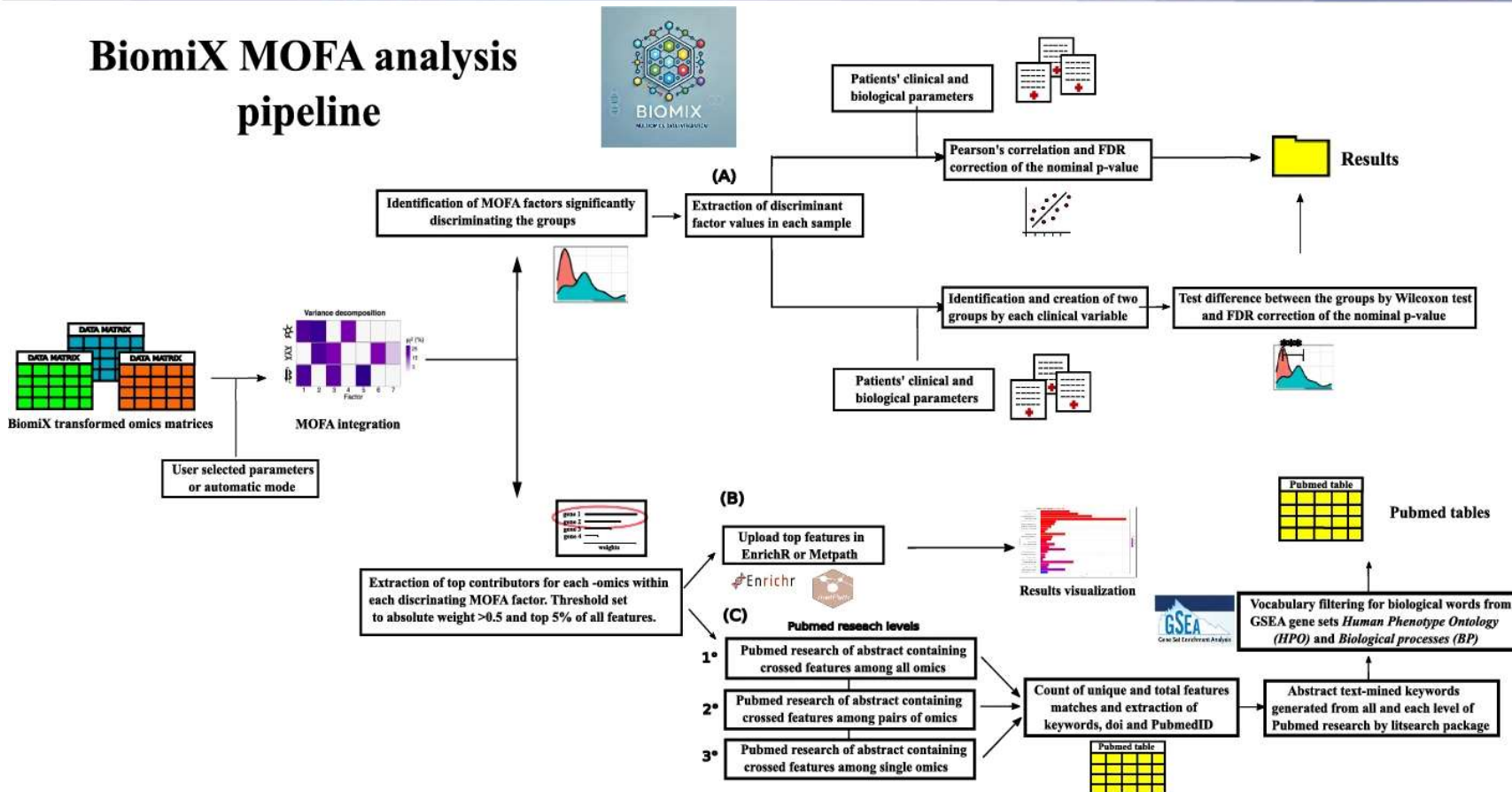
Unlike SNF: no iterative network diffusion.

Handles missing omics
Each pair of samples must share ≥ 1 measured omic





BiomiX MOFA analysis pipeline



What's new with BiomiX?

- Automatic selection MOFA/ DIABLO factors
- (stopping when new variance capture < 1%)
- Selection of factors separating the two conditions (semi-supervised)
- Automatic annotation of the latent factors
- - Correlation by clinical data
- - Pathway analysis
- - Bibliographic research

Iperi et al BMC bioinformatics (2025)



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MOFA tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



INTEGRATION OPTIONS

INTEGRATION
☒ Yes ☐ No

METHOD
☒ MOFA
☐ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS
N° factors (0 = auto)

Factor to explore

Omics overlay

Main interface

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General Metabolomics **MOFA & Diabolo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Advance option MOFA



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diablo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics **MOFA & Diablo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Converging speed, the slower the more accurate

All features contribution to a factor are ranked from 0 to 1. Score threshold to filter features for interpretation analysis

Advance option MOFA



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Biomix v2.5
Multiomics analysis pipeline



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

EGAS00001001746_metadata_CLL.xls

✓ 184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS

Condition (GROUP_1)

mutated

Control (GROUP_2)

unmutated

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

Select folder...

C:/Users/crist/Desktop/Biomix2.5

- 1) Load MTBLS7623_Metadata.tsv from Metadata folder.
- 2) Set the PTB vs HC comparison
- 3) Output folder in Biomix 3.0 folder



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OMICS INPUTS							
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Transcriptomics	☒	RNA	☐	Browse file... EGAS00001001746_transcriptomics.xls C:/Users/crist/Desktop/Biomix2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls
Input 2	☐	☐	Methylomics	☒	Methy	☐	Browse file... EGAS00001001746_methylomics.xls C:/Users/crist/Desktop/Biomix2.5/Methylomics/INPUT/EGAS00001001746_methylomics.xls
Input 3	☐	☐	Undefined	☐	Label input 3	☐	Browse file... No file selected

- 4) Load MTBLS7623_RNA_seq_OK.tsv from Transcriptomics folder.
- 5) Load MTBLS7623_positive_mode_annotated_HMDB.tsv from Metabolomics folder.
- 6) Click on integration to select the slot to integrate.



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diablo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

10

Factor to explore

1

Omics overlay

0

Select 10 factors

Advanced Options

▶ Run Biomix

Download JSON

Ready to run





1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Each omics has the positive and negative contributors for the “discriminating” factors

Example:

	A	B	C
1	weights	features	abs_weight
2	-0.50166	PLSCR1	0.501663
3	-0.50674	DYSF	0.506739
4	-0.50704	FBXO6	0.507038
5	-0.50839	MYOF	0.508391
6	-0.50894	LY6E	0.508945
7	-0.50907	RTP4	0.509069
8	-0.51414	LAMP3	0.514143
9	-0.51486	IFI44	0.514859
10	-0.51498	SECTM1	0.514976
11	-0.51513	SOCS1	0.515128
12	-0.51806	PARP14	0.518058
13	-0.52497	DHRS9	0.52497

Distribution_factors_contributions_PTB_vs...	17/09/2026 20:00	Adobe Acrobat D...	24 KB
METAB_Metabolomics_neg_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB
METAB_Metabolomics_pos_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB
model.hdf5	17/09/2026 20:00	File HDF5	1.228 KB
MOFA_explained_factors_10.tsv	17/09/2026 20:00	File TSV	1 KB
MOFA_explained_factors_rounded_10.tsv	17/09/2026 20:00	File TSV	1 KB
MOFA_explained_tot_10.tsv	17/09/2026 20:00	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	17/09/2026 20:00	File TSV	5 KB
MOFA_PTB_vs_HC_1.pdf	17/09/2026 20:00	Adobe Acrobat D...	361 KB
MOFA_SEPARATION_PTB_vs_HC_10_facto...	17/09/2026 20:00	File TSV	1 KB
RNA_Transcriptomics_neg_factor_1.tsv	17/09/2026 20:00	File TSV	3 KB
RNA_Transcriptomics_pos_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB



1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Variance explained by each factor within each omics

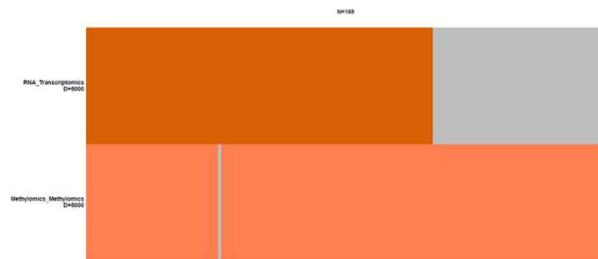
Latent factor value (10 factors) per each sample. Dimensional reduction-like.

P.adjusted value of the “discrimination” power of each factor among the two conditions

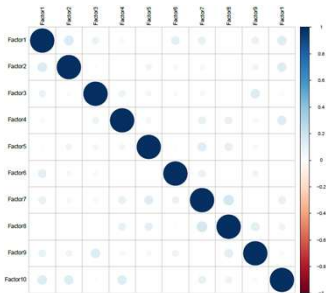
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	METAB_Metabolomics_neg_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB
	METAB_Metabolomics_pos_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB
	model.hdf5	17/09/2026 20:00	File HDF5	1.228 KB
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	MOFA_PTB_vs_HC_1.pdf	17/09/2026 20:00	Adobe Acrobat D...	361 KB
	MOFA_SEPARATION_PTB_vs_HC_10_facto...	17/09/2026 20:00	File TSV	1 KB
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	RNA_Transcriptomics_pos_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB



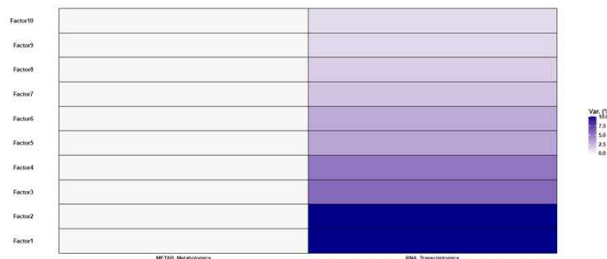
1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors



Sample distribution
per omics



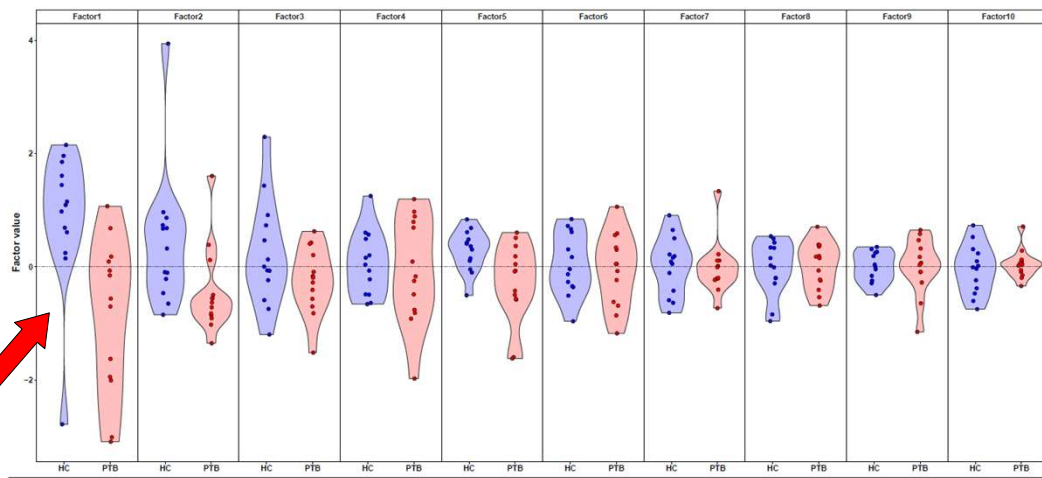
Factor co-
correlation test



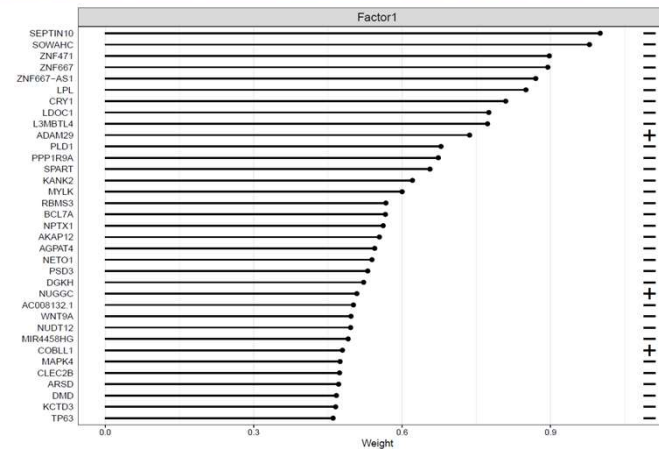
Variance explained per omics
and factor

Main pdf results including

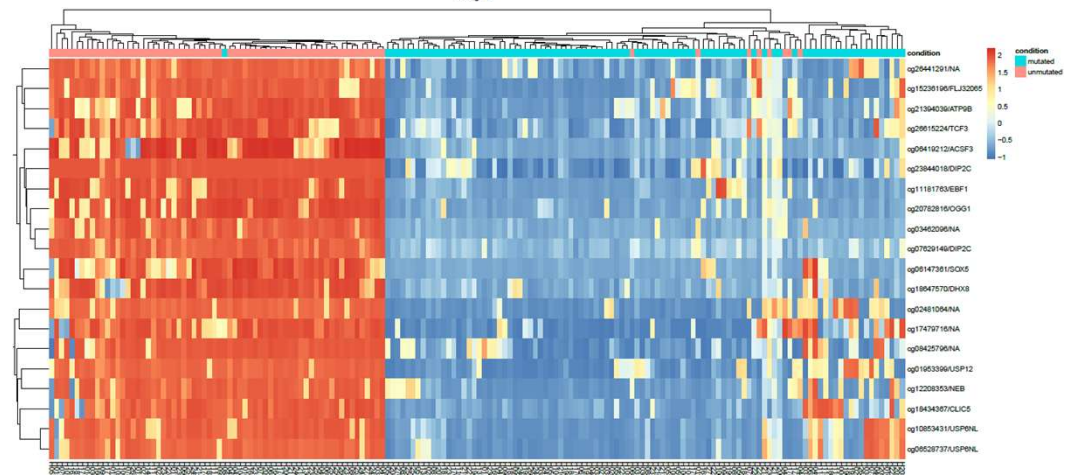
Distribution_factors_contributions_PTB_vs...	17/09/2026 20:00	Adobe Acrobat D...	24 KB
METAB_Metabolomics_neg_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB
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model.hdf5	17/09/2026 20:00	File HDF5	1.228 KB
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MOFA_explained_tot_10.tsv	17/09/2026 20:00	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	17/09/2026 20:00	File TSV	5 KB
MOFA_PTB_vs_HC_1.pdf	17/09/2026 20:00	Adobe Acrobat D...	361 KB
MOFA_SEPARATION_PTB_vs_HC_10_facto...	17/09/2026 20:00	File TSV	1 KB
RNA_Transcriptomics_neg_factor_1.tsv	17/09/2026 20:00	File TSV	3 KB
RNA_Transcriptomics_pos_factor_1.tsv	17/09/2026 20:00	File TSV	1 KB



Violin plot factor separation per condition



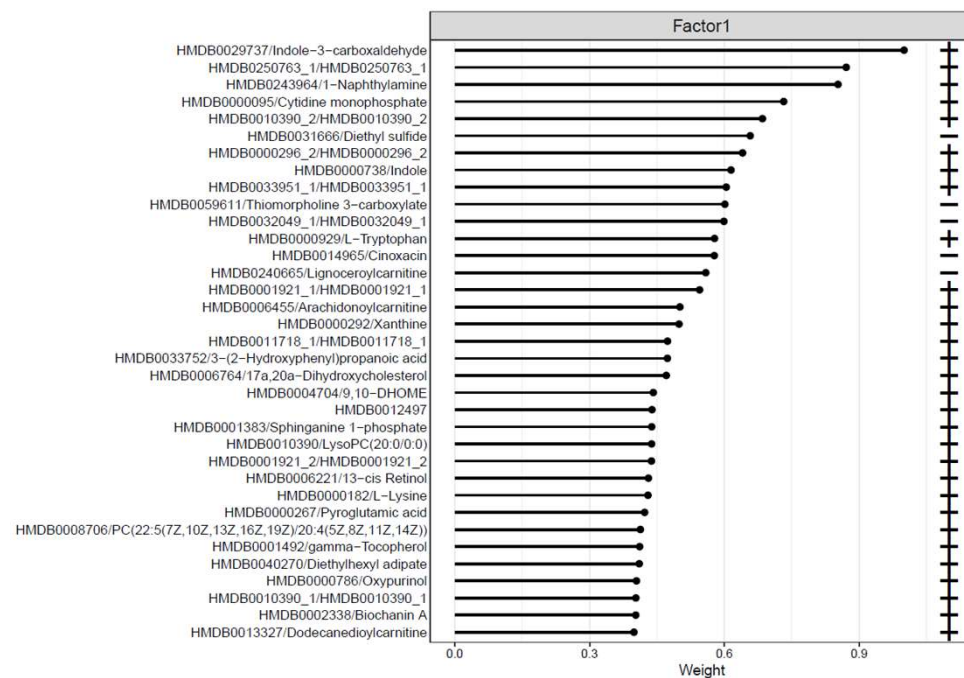
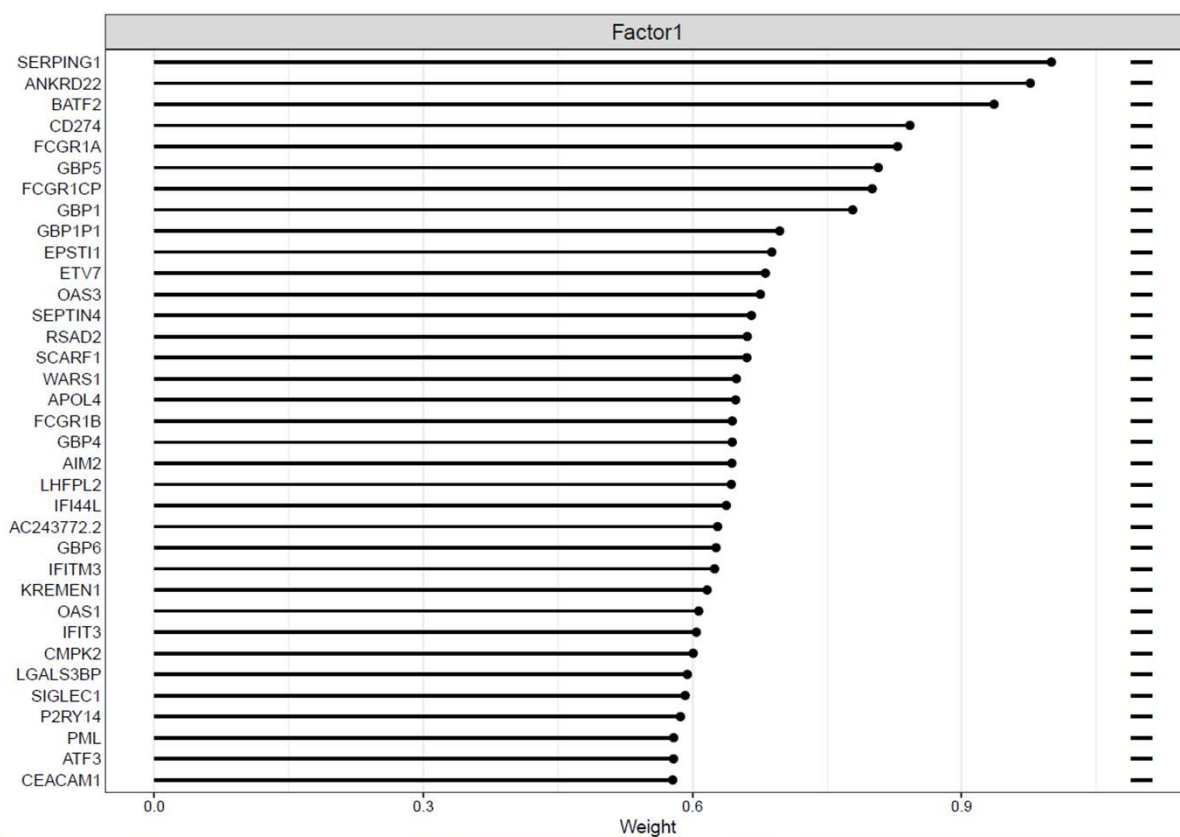
Top contributor weights.



Heatmap to features
factor per omics



1) Open output folder: Integration\OUTPUT\MOFA_PTB_vs_HC_10_factors





Nice, but what is behind
factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

FACTOR INTERPRETATION — USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type:

N° articles retrieved:

N° top Keywords extracted:

N° top contributors:

PATHWAY ANALYSIS

P.adj threshold:

N° pathways shown:

CLINICAL DATA INTEGRATION

Data types to correlate:

- ☒ Numerical metadata
- ☒ Binary metadata

Numerical clinical data file (optional): [MTBLS7623_clinical_numeric.xlsx](#)

Binary clinical data file (optional): [MTBLS7623_clinical_binary.xls](#)



Nice, but what is behind
factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

FACTOR INTERPRETATION — USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type:

N° articles retrieved:

N° top Keywords extracted:

N° top contributors:

PATHWAY ANALYSIS

P.adj threshold:

N° pathways shown:

CLINICAL DATA INTEGRATION

Data types to correlate:

- ☒ Numerical metadata
- ☒ Binary metadata

Numerical clinical data file (optional): [MTBLS7623_clinical_numeric.xlsx](#)

Binary clinical data file (optional): [MTBLS7623_clinical_binary.xls](#)

1) Re-run including
Numeric and binary
clinical data. These will
be correlated and tested
with the MOFA factors
calculated.

Are they associated?



Open:BiomiX2.5\Clinical_data\OUTPUT\ MOFA_PTB_vs_HC_10_factors



	A	B	C	D
1	clinic	means	pval	p.adj
2	GENDER	-0.56228	0.211556	0.211556
3	isolate	NA	1	X
4	factors	NA	1	X

	A	B	C	D
1	Numeric_c	p.value	Correlation	p.adj
2	AGE	0.013134	0.479741	0.039403
3	AvgSpotLe	0.037387	-0.41023	0.056081
4	Bases	0.502503	-0.13765	0.502503

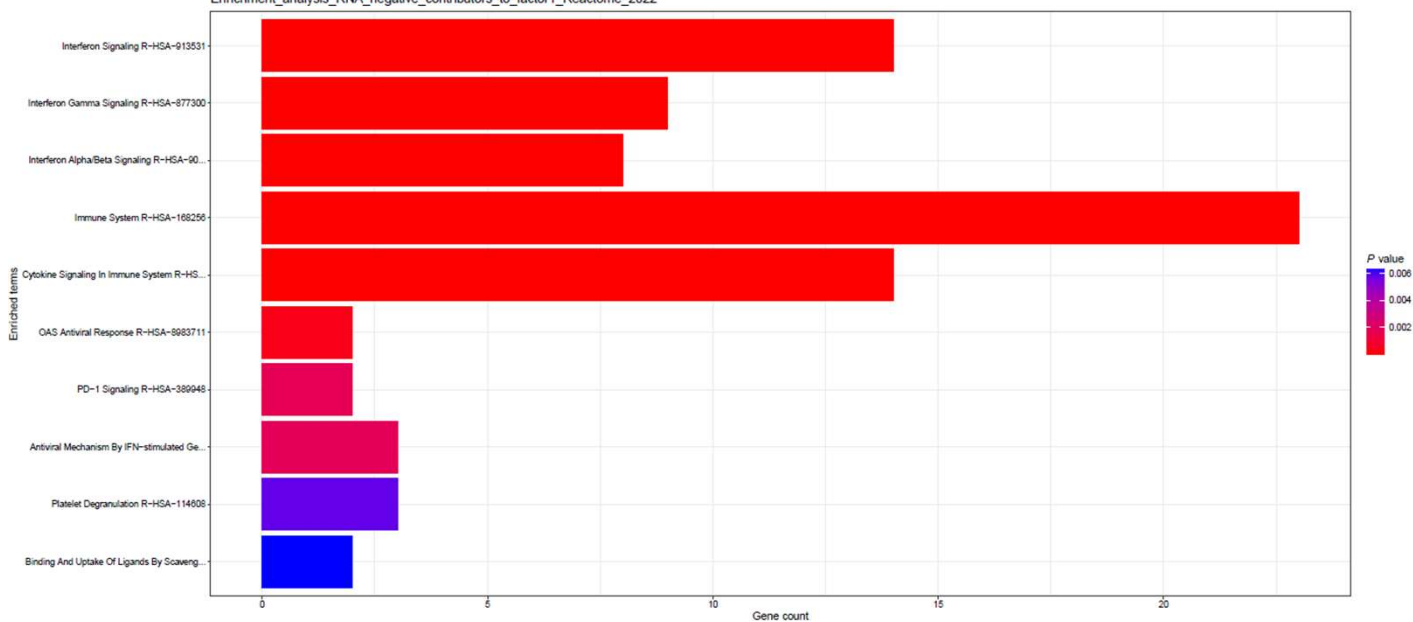
1) Conclusion:
Factor 1 is directly
associated to the IGHV
mutation measurement



Open:BiomiX2.5\Clinical_data\OUTPUT\ MOFA_PTB_vs_HC_10_factors\Factor_Pathways

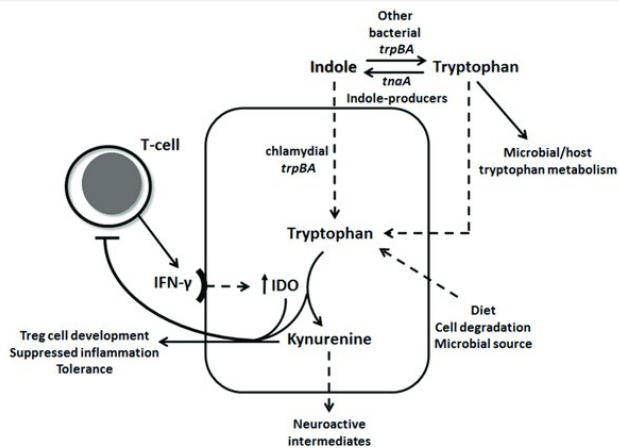


Enrichment_analysis_RNA_negative_contributors_to_factor1_Reactome_2022





1) Open output folder: Integration\OUTPUT\MOFA_PTB_vs_HC_10_factors



Enrichr

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Description No description available (59 genes)

Reactome Pathways 2024

- Interferon Signaling
- Interferon Gamma Signaling
- Interferon Alpha Beta Signaling
- Immune System
- Cytokine Signaling in Immune System

WikiPathways 2024 Human

- Type II Interferon Signaling WP619
- Immune Response To Tuberculosis WP4197
- Network Map Of SARS CoV 2 Signaling WP51
- Non Genomic Actions Of 1 25 Dihydroxyvita
- Cancer Immunotherapy By PD 1 Blockade V

BioPlanet 2019

- Interferon signaling
- Immune system signaling by interferons, int
- Interferon-gamma signaling pathway
- Immune system
- Interferon alpha/beta signaling

Ziklo et al Front Cell Infect Microbiol
. 2018 Jan 18;8:1. doi: 10.3389/fcimb.2018.00001



2026 Cristian Iperi, Jessica Gliozzo — CC BY 4.0



DIABLO tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



INTEGRATION OPTIONS

INTEGRATION
☒ Yes ☐ No

METHOD
☐ MOFA
☒ **Diablo**
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS
N° factors (0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

Main interface DIABLO

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics **MOFA & Diablo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations Convergence mode Contribution threshold

DIABLO PARAMETERS

Feature selection N° iterations Design matrix file (optional) Nessun file (opzionale)

Advance option DIABLO





🔍 FACTOR INTERPRETATION – USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type:

N° articles retrieved:

N° top Keywords extracted:

N° top contributors:

PATHWAY ANALYSIS

P.adj threshold:

N° pathways shown:

CLINICAL DATA INTEGRATION

Data types to correlate:

- ☒ Numerical metadata
- ☒ Binary metadata

Numerical clinical data file (optional): [EGAS00001001746_clinical_numeric.xlsx](#)

Binary clinical data file (optional): [EGAS00001001746_clinical_binary.xls](#)

**Don't forget the Clinical data
for the interpretation!**





1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

Example:

	A	B	C
1	features	weights	abs_weight
2	HMDB0006221/13-cis Retinol	-0.65571	0.655713
3	HMDB0006764/17a,20a-Dihydroxy	-0.68025	0.680252
4	HMDB0033951_1/HMDB0033951_	-0.68541	0.685413
5	HMDB0010386_1/HMDB0010386_	-0.69108	0.691075
6	HMDB0010390_2/HMDB0010390_	-0.72717	0.727168
7	HMDB0000296_2/HMDB0000296_	-0.72822	0.728217
8	HMDB0000095/Cytidine monophos	-0.73124	0.731243
9	HMDB0002815/LysoPC(18:1(9Z)/0:	-0.73646	0.73646
10	HMDB0000289/Uric acid	-0.76484	0.764836
11	HMDB0010390_1/HMDB0010390_	-0.76702	0.76702
12	HMDB0000738/Indole	-0.77417	0.774174
13	HMDB0010392/LysoPC(20:2(11Z,1	-0.78998	0.789981
14	HMDB0250763/D-Histidine	-0.81171	0.811713
15	HMDB0000929/L-Tryptophan	-0.81558	0.815577
16	HMDB0250763_1/HMDB0250763_	-0.8271	0.827098
17	HMDB0033951/Racemethionine	-0.8338	0.833801
18	HMDB0243964/1-Naphthylamine	-0.85224	0.852243
19	HMDB0029737/Indole-3-carboxald	-0.98238	0.982381
20	HMDB0030396/(Å*)-Tryptophan	-0.98712	0.987124
21	HMDB0000159/L-Phenylalanine	-1	1

Each omics has the positive and negative contributors for the “discriminating” factors

Clinical_correlation	18/09/2026 00:35	Cartella di file	
Factor_Pathways	18/09/2026 00:35	Cartella di file	
Factors_Associated_Articles	18/09/2026 00:37	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	18/09/2026 00:34	File TSV	10 KB
DIABLO_MATRIX_PTB_vs_HC_10_factors.tsv	18/09/2026 00:34	File TSV	1 KB
DIABLO_PTB_vs_HC.pdf	18/09/2026 00:34	Adobe Acrobat D...	533 KB
DIABLO_SEPARATION_PTB_vs_HC_factors....	18/09/2026 00:34	File TSV	2 KB
Distribution_factors_contributions_PTB_vs...	18/09/2026 00:34	Adobe Acrobat D...	19 KB
METAB_Metabolomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	2 KB
METAB_Metabolomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	1 KB
RNA_Transcriptomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	6 KB
RNA_Transcriptomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	5 KB



1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

Latent factor value (10 factors) per each sample. Dimensional reduction-like.

Variance explained by each factor within each omics

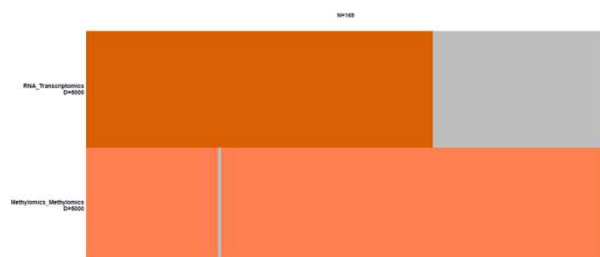
P.adjusted value of the “discrimination” power of each factor among the two conditions

Clinical_correlation	18/09/2026 00:35	Cartella di file	
Factor_Pathways	18/09/2026 00:35	Cartella di file	
Factors_Associated_Articles	18/09/2026 00:37	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	18/09/2026 00:34	File TSV	10 KB
DIABLO_MATRIX_PTB_vs_HC_10_factors.tsv	18/09/2026 00:34	File TSV	1 KB
DIABLO_PTB_vs_HC.pdf	18/09/2026 00:34	Adobe Acrobat D...	533 KB
DIABLO_SEPARATION_PTB_vs_HC_factors....	18/09/2026 00:34	File TSV	2 KB
Distribution_factors_contributions_PTB_vs...	18/09/2026 00:34	Adobe Acrobat D...	19 KB
METAB_Metabolomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	2 KB
METAB_Metabolomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	1 KB
RNA_Transcriptomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	6 KB
RNA_Transcriptomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	5 KB

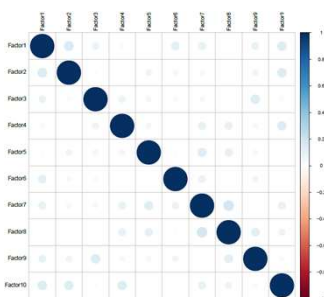


1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors

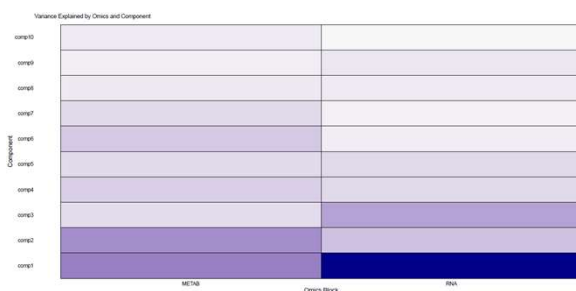
Main pdf results including



Sample distribution per omics

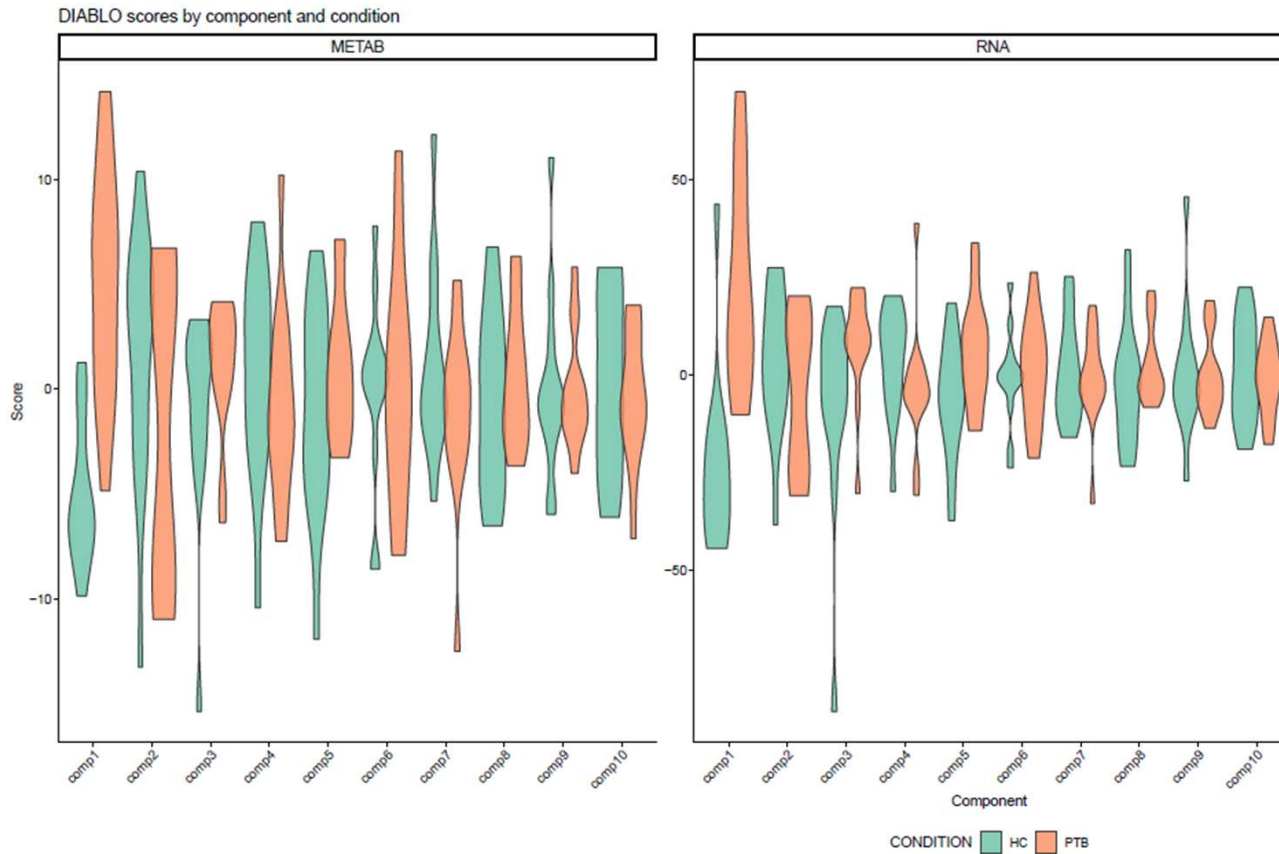


Factor co-correlation test



Variance explained per omics and factor

Clinical_correlation	18/09/2026 00:35	Cartella di file	
Factor_Pathways	18/09/2026 00:35	Cartella di file	
Factors_Associated_Articles	18/09/2026 00:37	Cartella di file	
DIABLO_FACTORS_WEIGHTS.tsv	18/09/2026 00:34	File TSV	10 KB
DIABLO_MATRIX_PTB_vs_HC_10_factors.tsv	18/09/2026 00:34	File TSV	1 KB
DIABLO_PTB_vs_HC.pdf	18/09/2026 00:34	Adobe Acrobat D...	533 KB
DIABLO_SEPARATION_PTB_vs_HC_factors....	18/09/2026 00:34	File TSV	2 KB
Distribution_factors_contributions_PTB_vs...	18/09/2026 00:34	Adobe Acrobat D...	19 KB
METAB_Metabolomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	2 KB
METAB_Metabolomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	1 KB
RNA_Transcriptomics_neg_factor_1.tsv	18/09/2026 00:34	File TSV	6 KB
RNA_Transcriptomics_pos_factor_1.tsv	18/09/2026 00:34	File TSV	5 KB

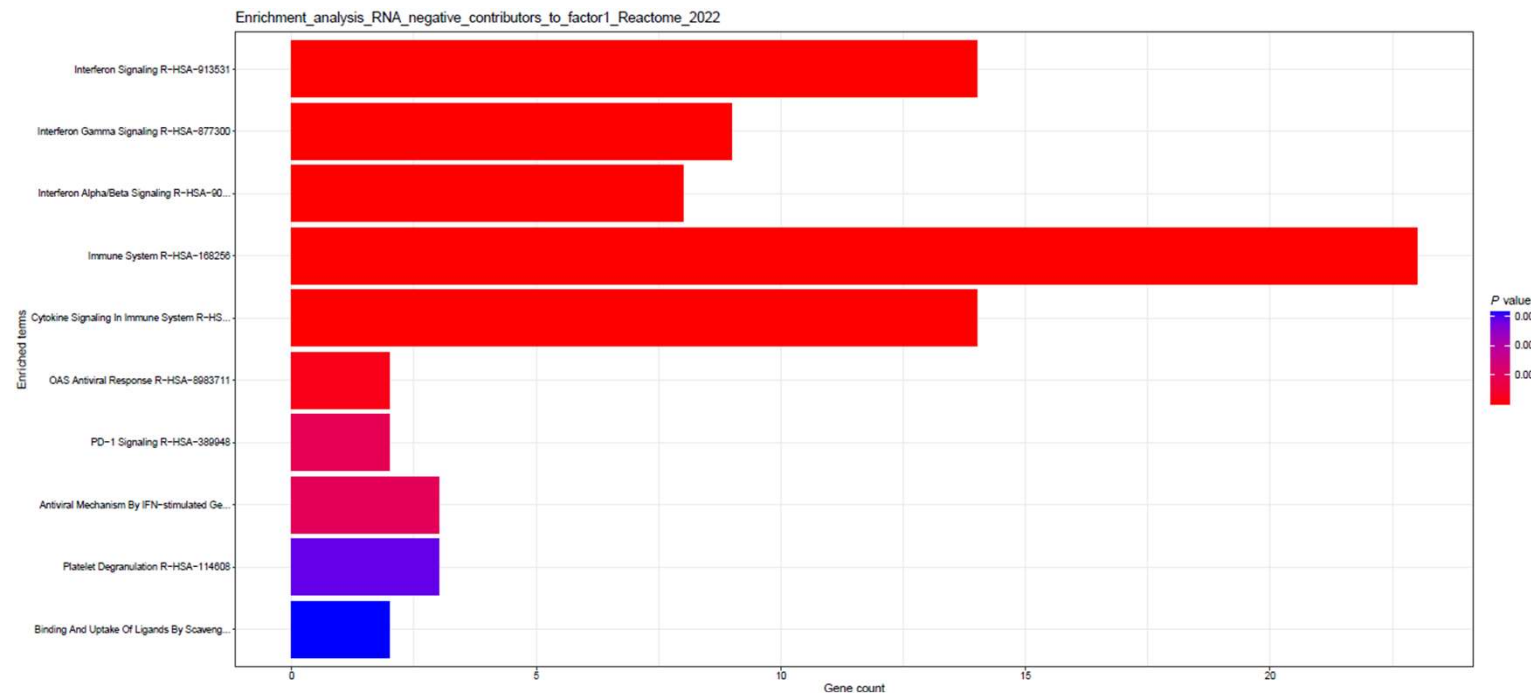


1) Factor 1 is discriminating patients.

Which biological processes are behind it?

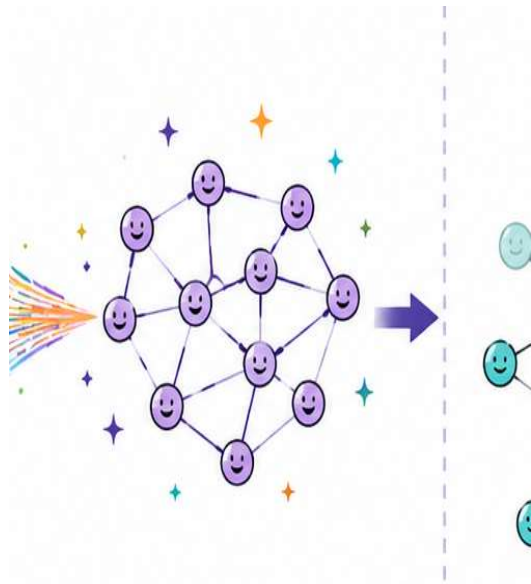


1) Open output folder: Integration\OUTPUT\DIABLO_mutated_vs_unmutated_10_factors\Factor_Pathways



1) HC patients are associated to higher values in factor 1 → Positive contributors

1) PTB patients are associated to Lower values in factor 1 → Negative contributors



SNF & NEMO Tutorials

Network-based multi-omics approaches



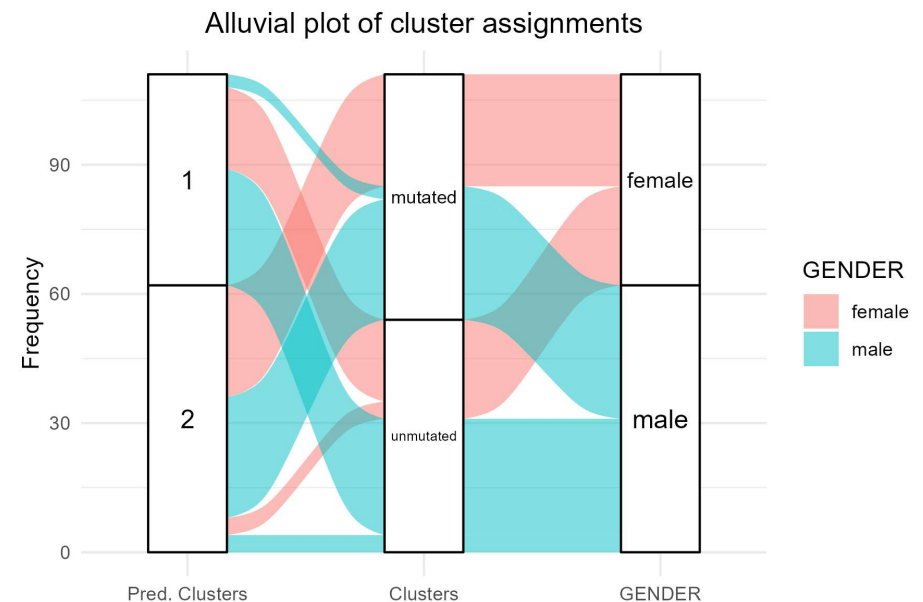


Number of clusters optimization SNF/NEMO

- Optimal number of clusters estimation: **eigengap/modified eigegap method**
- Internal clustering **validation metrics** (i.e. *"Mean Silhouette Coefficient"*, *"Dunn Index"*, *"Entropy"* and *"Calinski and Harabasz Index"*)

Implemented interpretation methods SNF/NEMO

- Similarities heatmap for single omics and integrated matrix
- Network (graph) for single omics and integrated network
- Post-hoc feature prioritization based on NMI
- Variables enrichment (chi-squared test for categorical, Kruskal-Wallis for continuous variables)
- Survival analysis (log-rank test and Kaplan-Meier plots)
- Alluvial plot comparing clustering, CONDITION (if any), variables of interest





1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

MTBLS7623_Metadata.tsv

39 samples · 3 conditions: HC, PTB, PTB_DM

2. COMPARISON GROUPS

Condition (GROUP_1)

PTB

Control (GROUP_2)

HC

3. OUTPUT FOLDER

Running in Docker: results will be saved to the folder you mounted on your machine via -v <your-folder>:/shared

/shared

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION (=RUN SINGLE OMICS BEFORE)	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Transcriptomics	☒	RNA	☐	Browse file... No file selected
Input 2	☐	☐	Metabolomics	☒	METAB	☐	Browse file... MTBLS7623_positive_mode_unannotated.tsv /shared/MTBLS7623_positive_mode_unannotated.tsv





INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☐ MOFA
☐ Diablo
☒ SNF
☐ NEMO
☐ MintTea

PARAMETERS

 Method-specific parameters for this integration method are in **Advanced Options**.

 **Advanced Options**

 **Run BiomiX**

 **Download JSON**

 **Ready to run**



Advanced Options SNF

Model:

- Number of nearest neighbours
- Variance (controls decay of similarity with distance)
- Number of message-passing iterations

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)

Setup | **Advanced Options** | QC & Imputation | JSON Preview

General | Metabolomics | MOFA & Diabolo | **SNF / NEMO** | MintTea | Metadata Filtering

SNF PARAMETERS

N° neighbors: 15 | Variance affinity matrix: 0.5 | N° iterations: 20

NEMO PARAMETERS

N° neighbors: 5 | Only common samples: TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test: AGE GENDER | Columns for Time (add .time) and survival column (add .event): | Condition column: CONDITION

CLUSTERING AND FEATURE OPTIONS

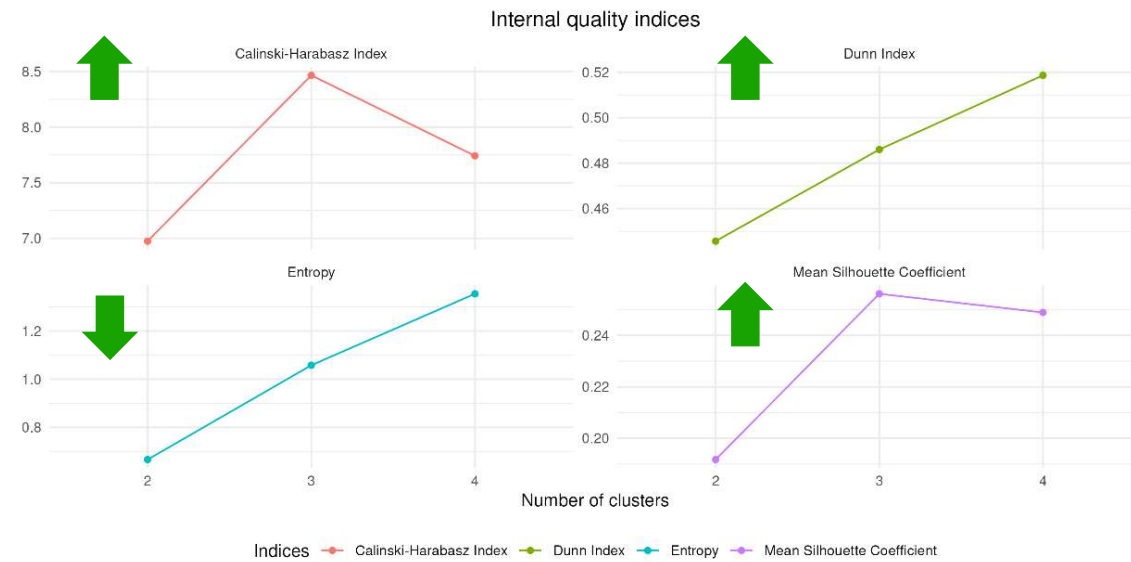
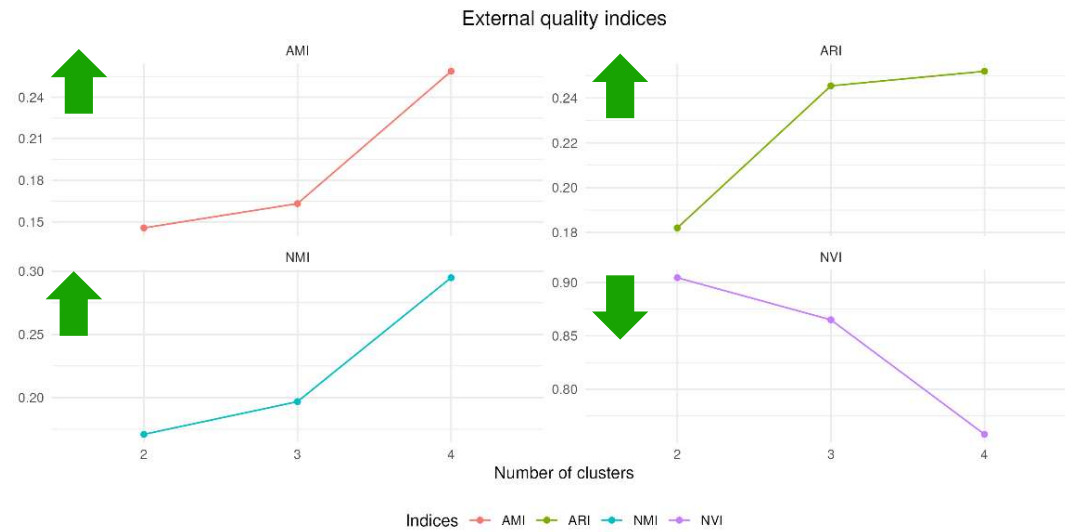
Max cluster range: 6 | N° variables in heatmap: 20 | Max N° features SNF / NEMO: 5000 | Scaling the SNF/NEMO data: TRUE





Let's explore the results!





Metric	What it measures	Better when...
Silhouette Coefficient	How well-separated and compact clusters are	Higher
Dunn Index	Ratio of min inter-cluster distance to max intra-cluster distance	Higher
Calinski-Harabasz Index	Ratio of between-cluster to within-cluster dispersion	Higher
Entropy	Purity of clusters w.r.t. class labels	Lower
AMI	Mutual info between two clusterings, adjusted for chance	Higher
ARI	Similarity between two clusterings, adjusted for chance	Higher
NMI	Normalized shared information between two clusterings	Higher
NVI	Normalized distance (disagreement) between two clusterings	Lower

Integration\OUTPUT\SNF_PTB_vs_HC\clustering_metrics





Switch in the main interface to NEMO in the “Integration Options”
(we can avoid single-omics analysis since already run for SNF)

INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

☐ MOFA

☐ Diablo

☐ SNF

☒ NEMO

☐ MintTea

PARAMETERS

i Method-specific parameters for this integration method are in [Advanced Options](#).



⊙ Setup **Advanced Options** 📄 QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo **SNF / NEMO** MintTea Metadata Filtering

SNF PARAMETERS

N° neighbors: 10 Variance affinity matrix: 0.5 N° iterations: 20

NEMO PARAMETERS

N° neighbors: 15 Only common samples: TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test: AGE GENDER Columns for Time (add.time) and survival column (add.event): Condition column: CONDITION

CLUSTERING AND FEATURE OPTIONS

Max cluster range: 6 N° variables in heatmap: 20 Max N° features SNF / NEMO: 5000 Scaling the SNF/NEMO data: TRUE

Advanced Options NEMO

Model:

- Number of nearest neighbours
- Consider only samples having all omics available

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)



Let's explore the results!





Issues & suggestions

BiomiX is an actively developed research project; if you encounter any problem or have suggestions, please open an issue on GitHub or contact the authors.

GitHub issues page link: <https://github.com/BiomiX-consortium/BiomiX2.5/issues>

Authors contacts:

Cristian Iperi for issues with single-omics pipelines, MOFA, DIABLO, others: **cristian.iperi@uzh.ch**

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